

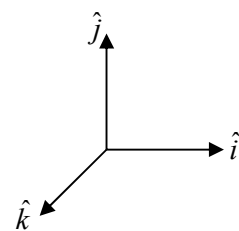
Chapter 1

Tools of Quantum Mechanics

1. Analogy Linear Vector Space

Three-Dimensional Vector Space

A vector in three-dimensional space can be denoted by $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ which have direction and magnitude where the real numbers a_1, a_2, a_3 are responsible of magnitude and unit vector $\hat{i}, \hat{j}, \hat{k}$ are responsible for component along three perpendicular direction.



In another word we can say that vector $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ can be represented in orthonormal basis of unit vector $\hat{i}, \hat{j}, \hat{k}$. The magnitude of vector $\vec{A}$ is denoted by $|\vec{A}|$, which is equivalent to $|\vec{A}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$.

In this space we define another vector $\vec{B}$ merely changing scalar a_1, a_2, a_3 to b_1, b_2, b_3 and we find another vector $\vec{B} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$. So we can construct infinite number of vectors in basis of unit

vector $\hat{i}, \hat{j}, \hat{k}$ which represent three dimensional space. These vectors are example of linear vector spaces.

The vectors in three-dimensional space have given property.

A. Vector Addition

(a) Closer Relation: If vector $\vec{A}$ defined by $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ represented in basis $\hat{i}, \hat{j}, \hat{k}$ and another vector $\vec{B}$ defined in same basis $\hat{i}, \hat{j}, \hat{k}$ then addition of $\vec{A}$ and $\vec{B}$ will produce vector $\vec{C}$.

Where vector $\vec{C} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ can be represent in same basis vector $\hat{i}, \hat{j}, \hat{k}$. So $\vec{A} + \vec{B} = \vec{C}$ where $c_1 = a_1 + b_1, c_2 = a_2 + b_2$ and $c_3 = a_3 + b_3$

(b) Commutative Law: $\vec{A} + \vec{B} = \vec{B} + \vec{A} = \vec{C}$

(c) Associative law: $(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$

(d) Existence of Null Vector: A null vector $\vec{O}$ must be part of vector spaces which after vector addition to vector $\vec{A}$ give same vector $\vec{A}$. In very general view Null vector is definition of origin.

$$\vec{A} + \vec{O} = \vec{A}$$

B. Scalar multiplication to a vector

Linear Vector spaces also constitute a scalar number $a, b, c, \dots$ which is real number which follow following property with vectors. $a\vec{A}$ means the vector is scaled up or down in magnitude by value a .

$$a(\vec{A} + \vec{B}) = a\vec{A} + a\vec{B}$$

$$(a + b)\vec{A} = a\vec{A} + b\vec{A}$$

$$ab\vec{A} = ba\vec{A}$$

$$0\vec{A} = \vec{O}$$

C. Scalar Product

The scalar product between two vectors $\vec{A}$ and $\vec{B}$ is denoted by $\vec{A} \cdot \vec{B}$ where $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$ which is equivalent to projection of $\vec{A}$ on vector $\vec{B}$ multiply by magnitude of vector $\vec{B}$. Hence vector $\hat{i}, \hat{j}$ and $\hat{k}$ are orthonormal basis vector where $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{B} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then $\vec{A} \cdot \vec{B} = a_1b_1 + a_2b_2 + a_3b_3$.

The property of scalar product

(a) Scalar Products Are Commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$

(b) Homogeneous Under Scaling Variable: $(a\vec{A}) \cdot \vec{B} = a(\vec{A} \cdot \vec{B}) = \vec{A} \cdot (a\vec{B})$

(c) Distributive Property: $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$

(d) Normal Vector: The magnitude of vector $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ is also known as norm of vector can be defined by $|\vec{A}| = \sqrt{(\vec{A} \cdot \vec{A})} = \sqrt{a_1^2 + a_2^2 + a_3^2}$. If we find norm or magnitude of any vector then

we can normalize vector and treat them as unit vector $\hat{n}$, where $\hat{n} = \frac{\vec{A}}{|\vec{A}|} = \frac{\vec{A}}{\sqrt{\vec{A} \cdot \vec{A}}}$

(e) Orthogonal Vector: Two vectors $\vec{A}$ and $\vec{B}$ is said to be orthogonal vector if $\vec{A} \cdot \vec{B} = 0$

D. Triangle Inequality

For given two vectors $\vec{A}$ and $\vec{B}$ $|\vec{A} + \vec{B}| \leq |\vec{A}| + |\vec{B}|$

E. Cauchy-Schwarz Inequality

For given two vectors $\vec{A}$ and $\vec{B}$ $|\vec{A} \cdot \vec{B}| \leq |\vec{A}| \cdot |\vec{B}|$

If $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{B} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then according to Cauchy-schwarz inequality

$$\sqrt{a_1b_1 + a_2b_2 + a_3b_3} \leq \left(\sqrt{a_1^2 + a_2^2 + a_3^2} \right) \left(\sqrt{b_1^2 + b_2^2 + b_3^2} \right)$$

Use of orthonormal basis

If any vector that vector $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ can be represented in orthonormal basis of unit vector $\hat{i}, \hat{j}, \hat{k}$.

Which means the basis vector are treated as unit vectors or normal vectors i.e. $\hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1$ and $\hat{k} \cdot \hat{k} = 1$ as well as they are treated as orthogonal vectors i.e. $\hat{i} \cdot \hat{j} = 0, \hat{j} \cdot \hat{k} = 0$ and $\hat{k} \cdot \hat{i} = 0$.

Due to orthonormal property of basis vectors we can easily defined scalar quantity a_1, a_2 and a_3 by property $a_1 = \hat{i} \cdot \vec{A}$, $a_2 = \hat{j} \cdot \vec{A}$, $a_3 = \hat{k} \cdot \vec{A}$ so vector $\vec{A}$ can be redefined as

$$\vec{A} = (\hat{i} \cdot \vec{A})\hat{i} + (\hat{j} \cdot \vec{A})\hat{j} + (\hat{k} \cdot \vec{A})\hat{k}.$$

In the quantum mechanics these properties of orthonormal basis have huge role to define quantum mechanical state of system.

All these above mention analogies can be beautifully explained by linear vector algebra.

Chapter 1

Tools of Quantum Mechanics

2. Hilbert Space and Dirac Notation

The mathematical concept of a **Hilbert space**, named after David Hilbert, generalizes the notion of Euclidian space. It extends the methods of linear vector algebra and calculus from the two-dimensional Euclidean plane and three-dimensional Euclidean space to spaces with any finite or infinite number of dimensions. A Hilbert space is an abstract vector space possessing the structure of an inner product that allows length and angle to be measured.

Hilbert space H have all property consists of linear vector space. It consists of collection of vector or function $\phi, \xi, \psi \dots$ represented in Greek letters and scalar numbers $a, b, c \dots$. The functions $\phi, \xi, \psi \dots$ are complex function so their complex conjugate can be represented as $\phi^*, \xi^*, \psi^* \dots$ belong to dual space of Hilbert space H . In quantum mechanics $\phi, \xi, \psi \dots$ also said to be state of system. $\phi, \xi, \psi \dots$ and $\phi^*, \xi^*, \psi^* \dots$ are function of dynamical variable or generalized coordinate of classical system. The scalar numbers $a, b, c \dots$ are also complex numbers and also belong to dual space.

The function belongs to Hilbert space have followed property.

A. Addition Rule

(a) Closer Relation: If $\psi(x)$ and $\phi(x)$ are belongs to Hilbert Space H then their addition $\phi(x) + \psi(x) = \xi(x)$. $\xi(x)$ will also belong to same Hilbert space H .

(b) Commutation $\phi(x) + \psi(x) = \psi(x) + \phi(x)$

(c) Associative $(\phi(x) + \psi(x)) + \xi(x) = \phi(x) + (\psi(x) + \xi(x))$

(d) Existence of Null vector there exists a vector or function represented as O which addition to any function $\psi(x)$ will produce same vector i.e. $\psi(x) + O = \psi(x)$

(e) Existence of inverse: There must be existence of inverse vector of $\psi(x)$ which addition on $\psi(x)$ will produce null vector, i.e. $\psi(x) + (-\psi(x)) = O$. Which means $-\psi(x)$ is inverse of $\psi(x)$.

B. Scalar Multiplication

(a) The product of a scalar with a vector gives another vector. In general, if ϕ and ψ are two vectors of the space, any linear combination $a\phi + b\psi$ is also a vector of the space, a and b being scalars.

(b) Distributive with Respect to Addition

$$(a + b)\psi(x) = a\psi(x) + b\psi(x) \text{ and } a(\psi(x) + \phi(x)) = a\psi(x) + a\phi(x)$$

(c) Associativity with Respect to Multiplication of Scalars

$$a(b\psi(x)) = ab\psi(x)$$

(d) For each element ψ there must exist a unitary scalar I and a zero scalar O such that

$$I\psi = \psi \text{ and } 0\psi = O$$

Scalar Product: If two function $\phi(x)$ and $\psi(x)$ belongs to same Hilbert space H where x is dynamical variable of system then scalar product can be denoted by (ϕ, ψ) . The value of scalar product is

$$(\phi, \psi) = \int_{-\infty}^{\infty} \phi^*(x) \psi(x) dx < \infty. \text{ Famously it is also named as inner product. The inner product } (\phi, \psi) \text{ must be finite number.}$$

Suppose If two function $\phi(x, y)$ and $\psi(x, y)$ belongs to Hilbert space where dynamical variables

$$\text{are } x, y \text{ then scalar product is defined as } (\phi, \psi) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi^*(x, y) \psi(x, y) dx dy < \infty.$$

Similarly, if dynamical variables are spherical polar coordinate r, θ, ϕ then

$$(\phi, \psi) = \int_0^\infty \int_0^\pi \int_0^{2\pi} \phi^*(r, \theta, \phi) \psi(r, \theta, \phi) r^2 \sin \theta dr d\theta d\phi < \infty$$

Properties of Scalar Product

(a) $(\phi, \psi)^* = (\psi, \phi)$

(b) $(\psi, a_1 \phi_1 + a_2 \phi_2) = a_1 (\psi, \phi_1) + a_2 (\psi, \phi_2)$

(c) $(a_1 \phi_1 + a_2 \phi_2, \psi) = a_1^* (\phi_1, \psi) + a_2^* (\phi_2, \psi)$

(d) $(a_1 \phi_1 + a_2 \phi_2, b_1 \psi_1 + b_2 \psi_2) = a_1^* b_1 (\phi_1, \psi_1) + a_1^* b_2 (\phi_1, \psi_2) + a_2^* b_1 (\phi_2, \psi_1) + a_2^* b_2 (\phi_2, \psi_2)$

Square Integrable Function: The function ψ belongs to Hilbert space H with dynamical variable x . Then ψ must be square integrable. The condition for square integrability is

$$(\psi, \psi) = \int_{-\infty}^{+\infty} \psi^* \psi dx = \alpha = \int_{-\infty}^{+\infty} |\psi|^2 dx = \alpha,$$

where α is positive constant. So, the norms of function ψ is given by $N = \sqrt{(\psi, \psi)} = \sqrt{\alpha}$

Normalized Function: If norm of any function ψ belong to Hilbert space H is one then function is said to be normalized. Any square integrable function can be normalized when it is divided by its norm which is also known as normalization constant $A = \frac{1}{N} = \frac{1}{\sqrt{(\psi, \psi)}}$. In general normalization

condition is $(\psi, \psi) = \int_{-\infty}^{+\infty} \psi^* \psi dx = 1 \Rightarrow \int_{-\infty}^{+\infty} |\psi|^2 dx = 1.$

The normalized function can be treated as unit vector.

Orthogonal Function: If two function ϕ and ψ belong to same Hilbert space H . They are said to be orthogonal if scalar product between that function will vanish ie

$$(\psi, \phi) = 0 \Rightarrow \int_{-\infty}^{+\infty} \psi^*(x) \phi(x) dx = 0$$

Orthonormal Function: If two function ϕ_i and ϕ_j belong to same Hilbert space H . They are said

to be orthonormal if $(\phi_i, \phi_j) = \int_{-\infty}^{+\infty} \phi_i(x) \phi_j(x) dx = \delta_{ij} = \begin{cases} 1, i=j \\ 0, i \neq j \end{cases}$

Chapter 1

Tools of Quantum Mechanics

3. Dirac Notation

The function belongs to Hilbert space H can be represented in term of Dirac notation. It is abstract notation can be efficiently used in quantum mechanics.

If function $\psi(x)$ belongs to Hilbert space H with dynamical variable x then it can be represented as ket vector as $|\psi\rangle$ pronounced as ψ (shai) ket. Similarly, function $\phi(x)$ can be represented as $|\phi\rangle$ and pronounced as ϕ (phai) ket. The $\psi^*(x)$ belongs to dual space of can be represented as Bra vector $\langle\psi|$ pronounced as ψ (shai) Bra. Similarly, function $\phi^*(x)$ can be represented as $\langle\phi|$ and pronounced as ϕ (phai) bra.

Mathematically ket vector is equivalent to column vector as $|\psi\rangle \equiv \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

Bra vector is complex adjoint of ket vector i.e. Row vector represented as $\langle\psi| \equiv (a^* \quad b^* \quad c^*)$

For example if ket vector is given by $|\psi\rangle = \begin{pmatrix} 2 \\ -3 \\ 3+4i \end{pmatrix}$ then corresponding bra vector is

$$\langle\psi| = (2 \quad -3 \quad 3-4i)$$

A. Addition Rule

(a) Closer Relation If ket $|\psi\rangle$ and ket $|\phi\rangle$ are belongs to Hilbert Space H then their addition $|\phi\rangle + |\psi\rangle = |\xi\rangle$. Ket $|\xi\rangle$ will also belongs to same Hilbert space H .

(b) Commutation $|\phi\rangle + |\psi\rangle = |\psi\rangle + |\phi\rangle$

(c) Associative $(|\phi\rangle + |\psi\rangle) + |\xi\rangle = |\phi\rangle + (|\psi\rangle + |\xi\rangle)$

(d) Existence of Null vector there exists a vector or function represented as $|O\rangle$ which addition to any function $|\psi\rangle$ will produce same vector i.e. $|\psi\rangle + |O\rangle = |\psi\rangle$

(e) Existence of inverse There must be existence of inverse vector of Ket $|\psi\rangle$ which addition on $|\psi\rangle$ will produce null vector . i.e. $|\psi\rangle + (-|\psi\rangle) = |O\rangle$. Which means $-|\psi\rangle$ is inverse of $|\psi\rangle$.

B. Scalar Multiplication

(a) The product of a scalar with a vector gives another vector. In general, if $|\phi\rangle$ and $|\psi\rangle$ are two ketvectors of the space, any linear combination $a|\phi\rangle + b|\psi\rangle$ is also a vector of the space, a and b being scalars.

(b) Distributive with respect to addition

$$(a+b)|\psi\rangle = a|\psi\rangle + b|\psi\rangle \text{ and } a(|\psi\rangle + |\phi\rangle) = a|\psi\rangle + a|\phi\rangle$$

(c) Associativity with respect to multiplication of scalars

$$a(b|\psi\rangle) = ab|\psi\rangle$$

(d) For each element ψ there must exist a unitary scalar I and a zero scalar 0 such that

$$I|\psi\rangle = |\psi\rangle \text{ and } 0|\psi\rangle = |O\rangle$$

Scalar Product in Dirac Notation

Scalar Product: If two ket vectors $|\phi\rangle$ and $|\psi\rangle$ belongs to same Hilbert space H . where x is dynamical variable of system then scalar product can be denoted by $\langle\phi|\psi\rangle$. The value of scalar product is

$\langle \phi | \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x) \psi(x) dx < \infty$. Famously it is also named as inner product. The inner product

$\langle \phi | \psi \rangle$ must be finite number.

Properties of scalar product

(a) $\langle \phi | \psi \rangle^* = \langle \psi | \phi \rangle$

(b) $(|\psi\rangle, a_1|\phi_1\rangle + a_2|\phi_2\rangle) = a_1\langle\psi|\phi_1\rangle + a_2\langle\psi|\phi_2\rangle$

(c) $(a_1|\phi_1\rangle + a_2|\phi_2\rangle, |\psi\rangle) = a_1^*\langle\phi_1|\psi\rangle + a_2^*\langle\phi_2|\psi\rangle$

(d) $(a_1|\phi_1\rangle + a_2|\phi_2\rangle, b_1|\psi_1\rangle + b_2|\psi_2\rangle) = a_1^*b_1\langle\phi_1|\psi_1\rangle + a_1^*b_2\langle\phi_1|\psi_2\rangle + a_2^*b_1\langle\phi_2|\psi_1\rangle + a_2^*b_2\langle\phi_2|\psi_2\rangle$

Normalized Function: If norm of any ket vector $|\psi\rangle$ belong to Hilbert space H is one then function is said to be normalized. Any square integrable function can be normalized when it is divided by its norm which is also known as normalization constant $A = \frac{1}{N} = \frac{1}{\sqrt{\langle\psi|\psi\rangle}}$. In general

normalization condition is $\langle\psi|\psi\rangle = \int_{-\infty}^{+\infty} \psi^* \psi dx = 1 \Rightarrow \int_{-\infty}^{+\infty} |\psi|^2 dx = 1$.

Orthogonal Function: If two ket vector $|\phi\rangle$ and $|\psi\rangle$ belong to same Hilbert space H . They are said to be orthogonal if scalar product between that function will vanish ie

$$\langle\psi|\phi\rangle = 0 \Rightarrow \int_{-\infty}^{+\infty} \psi^*(x) \phi(x) dx = 0$$

Orthonormal Function: If two vectors $|\phi_i\rangle$ and $|\phi_j\rangle$ belong to same Hilbert space H . They are said

to be orthonormal if $\langle\phi_i|\phi_j\rangle = \delta_{i,j} \Rightarrow \int_{-\infty}^{+\infty} \phi_i(x) \phi_j(x) dx = \delta_{ij} = \begin{cases} 1, i=j \\ 0, i \neq j \end{cases}$, where $\delta_{i,j}$ is identified as

Kronecker delta

Forbidden State and Direct Product: If ket $|\phi\rangle$ and ket $|\psi\rangle$ belongs to same Hilbert space H then direct product $|\phi\rangle \otimes |\psi\rangle$ and $\langle\phi| \otimes \langle\psi|$ are meaningless, they are void and Nonsensical. If ket $|\phi\rangle$ and ket $|\psi\rangle$ belongs to different Hilbert space H then direct product $|\phi\rangle \otimes |\psi\rangle$ and $\langle\phi| \otimes \langle\psi|$ are identified as direct product.

Triangle Inequality: If ket $|\phi\rangle$ and ket $|\psi\rangle$ belongs to same Hilbert space H

$$\sqrt{\langle\psi + \phi|\psi + \phi\rangle} \leq \sqrt{\langle\psi|\psi\rangle} + \sqrt{\langle\phi|\phi\rangle}$$

Cauchy Swartz inequality If $|\phi\rangle$ and $|\psi\rangle$ belongs to same Hilbert space H then

$$|\langle\psi|\phi\rangle|^2 \leq (\langle\psi|\psi\rangle)(\langle\phi|\phi\rangle)$$

Proof: $0 \leq \|\phi\rangle - \lambda|\psi\rangle\|^2$ where $\lambda = \frac{\langle\phi|\psi\rangle}{\langle\psi|\psi\rangle}$ so $\lambda^* = \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle}$

$$\begin{aligned} 0 &\leq \|\phi\rangle - \lambda|\psi\rangle\|^2 \Rightarrow 0 \leq (\langle\phi - \lambda\psi|\phi - \lambda\psi\rangle) \\ &\Rightarrow 0 \leq \langle\phi|\phi\rangle - \lambda\langle\phi|\psi\rangle - \lambda^*\langle\psi|\phi\rangle + \lambda^*\lambda\langle\psi|\psi\rangle \\ &\Rightarrow 0 \leq \langle\phi|\phi\rangle - \frac{\langle\phi|\psi\rangle}{\langle\psi|\psi\rangle} \cdot \langle\phi|\psi\rangle - \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle} \langle\psi|\phi\rangle + \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle} \cdot \frac{\langle\phi|\psi\rangle}{\langle\psi|\psi\rangle} \langle\psi|\psi\rangle \\ &\Rightarrow \langle\phi|\phi\rangle \leq -\frac{\langle\phi|\psi\rangle}{\langle\psi|\psi\rangle} \cdot \langle\phi|\psi\rangle - \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle} \langle\psi|\phi\rangle + \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle} \cdot \frac{\langle\phi|\psi\rangle}{\langle\psi|\psi\rangle} \langle\psi|\psi\rangle \\ &\Rightarrow \langle\phi|\phi\rangle \leq -\frac{\langle\phi|\psi\rangle}{\langle\psi|\psi\rangle} \cdot \langle\phi|\psi\rangle - \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle} \langle\psi|\phi\rangle + \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle} \cdot \langle\phi|\psi\rangle \\ &\Rightarrow \langle\phi|\phi\rangle \leq -\frac{\langle\phi|\psi\rangle}{\langle\psi|\psi\rangle} \cdot \langle\phi|\psi\rangle \Rightarrow \langle\phi|\phi\rangle \langle\psi|\psi\rangle \leq -\langle\psi|\phi\rangle \cdot \langle\phi|\psi\rangle \\ &\Rightarrow \langle\phi|\phi\rangle \langle\psi|\psi\rangle \geq \langle\psi|\phi\rangle \cdot \langle\phi|\psi\rangle \Rightarrow |\langle\psi|\phi\rangle|^2 \leq (\langle\psi|\psi\rangle)(\langle\phi|\phi\rangle) \end{aligned}$$

Linear Independency

The vector $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle, \dots, |\phi_N\rangle$ are said to be linear independent if we find the coefficient $c_1, c_2, c_3, \dots, c_N$ which satisfy the equation. $c_1|\phi_1\rangle + c_2|\phi_2\rangle + c_3|\phi_3\rangle + \dots + c_N|\phi_N\rangle = 0$ and get the unique solution as

$c_1 = c_2 = c_3 = \dots = c_N = 0$ then $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle, \dots, |\phi_N\rangle$ is said to be linearly independent.

The *dimension* of a vector space is given by the *maximum number* of linearly independent vectors the space can have. For instance, if the maximum number of linearly independent vectors a space has is N $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle, \dots, |\phi_N\rangle$ this space is said to be N -dimensional. In this N -dimensional vector space, any vector $|\psi\rangle$ can be expanded as a linear combination i.e.

$$|\psi\rangle = c_1|\phi_1\rangle + c_2|\phi_2\rangle + c_3|\phi_3\rangle + \dots + c_N|\phi_N\rangle$$

Relation between orthogonal and linearly independent vectors

Every orthogonal vector is linearly independent but it is not necessary that every linear independent vector is orthogonal.

In quantum mechanics any vector $|\psi\rangle$ can be represented in orthonormal basis of vector $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle, \dots, |\phi_i\rangle, \dots, |\phi_j\rangle, \dots, |\phi_N\rangle$ as

$$|\psi\rangle = c_1|\phi_1\rangle + c_2|\phi_2\rangle + c_3|\phi_3\rangle + \dots + c_i|\phi_i\rangle + \dots + c_j|\phi_j\rangle + \dots + c_N|\phi_N\rangle \Rightarrow |\psi\rangle = \sum_{i=1}^N c_i|\phi_i\rangle$$

Hence it is given $\langle\phi_i|\phi_j\rangle = \delta_{i,j}$ so the term $c_i = \langle\phi_i|\psi\rangle$ and $c_j = \langle\phi_j|\psi\rangle$

So $|\psi\rangle = \sum_{i=1}^N c_i |\phi_i\rangle = \sum_{i=1}^N (\langle\phi_i|\psi\rangle) |\phi_i\rangle$ which means the coefficient c_i is scalar product between component basis vector $|\phi_i\rangle$ and vector $|\psi\rangle$. This trick can be only applied if vectors can be represented in orthonormal basis vector. Here basis vectors $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle, \dots, |\phi_i\rangle, \dots, |\phi_j\rangle, \dots, |\phi_N\rangle$ are analogically orthonormal unit vectors in Euclidian space.

Example: If $\psi(x) = \begin{cases} 0, & x < 0 \\ A \exp\left(-\frac{x}{a} + ikx\right), & x > 0 \end{cases}$ where a, k are positive constant, then find the value of A such that $\psi(x)$ is normalized.

Solution: $\psi(x) = \begin{cases} 0, & x < 0 \\ A \exp\left(-\frac{x}{a} + ikx\right), & x > 0 \end{cases}$ and $\psi^*(x) = \begin{cases} 0, & x < 0 \\ A \exp\left(-\frac{x}{a} - ikx\right), & x > 0 \end{cases}$

For normalization $(\psi, \psi) = 1 \Rightarrow \int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx = 1$

$$\int_{-\infty}^0 0 dx + \int_0^{\infty} A^* \exp\left(-\frac{x}{a} - ikx\right) A \exp\left(-\frac{x}{a} + ikx\right) dx = 1 \Rightarrow |A|^2 \int_0^{\infty} \exp\left(-\frac{2x}{a}\right) dx = 1$$

$$\Rightarrow A = \sqrt{\frac{2}{a}} \text{ so } \psi(x) = \begin{cases} 0, & x < 0 \\ \sqrt{\frac{2}{a}} \exp\left(-\frac{x}{a} + ikx\right), & x > 0 \end{cases} \text{ is normalized function.}$$

Example: Assume two function $\phi(x)$ and $\psi(x)$, which is defined as $\phi(x) = \begin{cases} a, & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$ and $\psi(x) = \begin{cases} bx + c, & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$. Find the condition on a, b, c (real numbers) such that $\phi(x)$ and $\psi(x)$ are orthonormal vector.

Solution: Condition such that $\phi(x)$ is normalized $(\phi, \phi) = \int_{-\infty}^{\infty} \phi^* \phi dx = 1 \Rightarrow \int_{-1}^1 a^2 dx = 1 \Rightarrow a = \frac{1}{\sqrt{2}}$

Condition such that $\phi(x)$ and $\psi(x)$ are orthogonal

$$(\phi, \psi) = 0 \Rightarrow \int_{-\infty}^{\infty} \phi^* \psi dx = 0 \Rightarrow \int_{-1}^1 \frac{1}{\sqrt{2}} (bx + c) dx = 0 \Rightarrow c = 0$$

Condition for $\psi(x)$ is normalized $(\psi, \psi) = 1 \Rightarrow \int_{-\infty}^{\infty} \psi^* \psi dx = 1 \Rightarrow \int_{-1}^1 (bx)^2 dx = 1 \Rightarrow b = \sqrt{\frac{3}{2}}$

So, $\phi(x) = \begin{cases} \frac{1}{\sqrt{2}}, & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$ and $\psi(x) = \begin{cases} \sqrt{\frac{3}{2}}x, & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$ are orthonormal functions.

Example: function $\phi(x) = \exp(-\alpha x^2)$, where $-\infty < x < +\infty$, and $\psi(x) = x \exp(-\alpha x^2)$ where $-\infty < x < \infty$ belong to Hilbert space with variable x

(a) Find norms of $\phi(x)$ and $\psi(x)$ write expression of normalized $\phi(x)$ and $\psi(x)$.

(b) Prove that $\phi(x)$ and $\psi(x)$ are orthogonal.

Use the integration $\int_0^{\infty} x^n \exp(-\beta x^2) dx = \frac{1}{2} \cdot \frac{1}{\beta^{\frac{n+1}{2}}} \left| \frac{n+1}{2} \right|$

Solution: (a) Norms of $\phi(x)$ is

$$N_1 = \sqrt{(\phi, \phi)}, (\phi, \phi) = \int_{-\infty}^{\infty} \phi^*(x) \phi(x) dx \Rightarrow \int_{-\infty}^{\infty} \exp(-2\alpha x^2) dx \Rightarrow 2 \times \frac{1}{2} \times \left(\frac{1}{2\alpha} \right)^{1/2} \left| \frac{1}{2} \right| = \left(\frac{1}{2\alpha} \right)^{1/2} \sqrt{\pi}$$

$$N_1 = \left(\frac{\pi}{2\alpha} \right)^{1/4}, \text{ so normalized constant } A_1 = \frac{1}{N_1} \Rightarrow A_1 = \left(\frac{2\alpha}{\pi} \right)^{1/4}$$

Normalized function $\phi(x) = \left(\frac{2\alpha}{\pi} \right)^{1/4} \exp(-\alpha x^2)$

Norms of $\psi(x)$ is $N_2 = \sqrt{(\psi, \psi)}$,

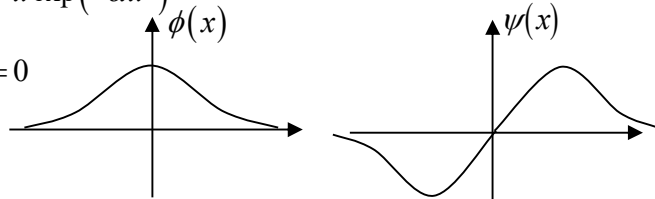
$$(\psi, \psi) = \int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx \Rightarrow \int_{-\infty}^{\infty} x^2 \exp(-2\alpha x^2) dx \Rightarrow 2 \times \frac{1}{2} \times \left(\frac{1}{2\alpha} \right)^{3/2} \left| \frac{3}{2} \right| = \left(\frac{1}{2\alpha} \right)^{3/2} \cdot \frac{1}{2} \cdot \sqrt{\pi}$$

$$N_2 = \left(\frac{1}{2} \right)^{1/2} \left(\frac{\pi^{1/4}}{(2\alpha)^{3/4}} \right) \text{ so normalized constant } A_2 = \frac{1}{N_2} \Rightarrow A_2 = 2^{1/2} \left(\frac{(2\alpha)^3}{\pi} \right)^{1/4}$$

Normalized function $\psi(x) = 2^{1/2} \left(\frac{(2\alpha)^3}{\pi} \right)^{1/4} x \exp(-\alpha x^2)$

(b) If ϕ and ψ are orthogonal, then $(\phi, \psi) = 0$

$$\Rightarrow A_1 \cdot A_2 \int_{-\infty}^{+\infty} e^{-\alpha x^2} x e^{-\alpha x^2} dx = 0$$



Example: (a) If $|\psi\rangle = A \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}$, Find the value of A such that $|\psi\rangle$ is normalized.

Solution: (a) If ket vector is $|\psi\rangle = A \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}$ then bra vector $\langle\psi| = A^* (1 \ -i \ 0)$

For normalization condition $\langle\psi|\psi\rangle = 1$

$$\Rightarrow A^* (1 \ -i \ 0) \cdot A \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = 1 \Rightarrow |A|^2 (1+1+0) = 1 \Rightarrow |A|^2 = \frac{1}{2} \Rightarrow A = \frac{1}{\sqrt{2}}$$

Example: If $|\psi\rangle = a_1|\phi_1\rangle + a_2|\phi_2\rangle$ it is also given $\langle\phi_i|\phi_j\rangle = \delta_{i,j}$ find $\langle\psi|\psi\rangle$ and $|\psi|^2$.

Solution: If ket $|\psi\rangle = a_1|\phi_1\rangle + a_2|\phi_2\rangle$

$$\langle\psi| = a_1^*\langle\phi_1| + a_2^*\langle\phi_2|$$

$$\langle\psi|\psi\rangle = a_1^*a_1\langle\phi_1|\phi_1\rangle + a_1^*a_2\langle\phi_1|\phi_2\rangle + a_2^*a_1\langle\phi_2|\phi_1\rangle + a_2^*a_2\langle\phi_2|\phi_2\rangle$$

Hence $\langle\phi_i|\phi_j\rangle = \delta_{i,j}$ so $\langle\phi_1|\phi_1\rangle = 1, \langle\phi_2|\phi_2\rangle = 1, \langle\phi_1|\phi_2\rangle = 0, \langle\phi_2|\phi_1\rangle = 0$

$$\langle\psi|\psi\rangle = a_1^*a_1 + a_2^*a_2 = |a_1|^2 + |a_2|^2 \text{ which is scalar number}$$

$$|\psi|^2 = \psi^*\psi, \psi = a_1\phi_1 + a_2\phi_2, \psi^* = a_1^*\phi_1^* + a_2^*\phi_2^*$$

$$\psi^*\psi = (a_1^*\phi_1^* + a_2^*\phi_2^*)(a_1\phi_1 + a_2\phi_2)$$

$$|\psi|^2 = a_1^*a_1\phi_1^*\phi_1 + a_1^*a_2\phi_1^*\phi_2 + a_2^*a_1\phi_2^*\phi_1 + a_2^*a_2\phi_2^*\phi_2 = |a_1|^2|\phi_1|^2 + a_1^*a_2\phi_1^*\phi_2 + a_2^*a_1\phi_2^*\phi_1 + |a_2|^2|\phi_2|^2$$

$$|\psi|^2 = |a_1|^2|\phi_1|^2 + |a_2|^2|\phi_2|^2 + 2\text{Re}(a_1^*a_2\phi_1^*\phi_2) \text{ which is a function}$$

Example: If $|\psi_1\rangle = a_1|\phi_1\rangle + a_2|\phi_2\rangle$ and $|\psi_2\rangle = b_1|\phi_1\rangle + b_2|\phi_2\rangle$ it is given $\langle\phi_i|\phi_j\rangle = \delta_{i,j}$ find the condition such that $|\psi_1\rangle$ and $|\psi_2\rangle$ are orthonormal.

Solution: Condition such that $|\psi_1\rangle$ is normalized $\langle\psi_1|\psi_1\rangle = 1$

$$\langle\psi_1|\psi_1\rangle = 1 \Rightarrow a_1^*a_1\langle\phi_1|\phi_1\rangle + a_2^*a_2\langle\phi_2|\phi_2\rangle + a_1^*a_2\langle\phi_1|\phi_2\rangle + a_2^*a_1\langle\phi_2|\phi_1\rangle = 1 \text{ if } \langle\phi_i|\phi_j\rangle = \delta_{i,j}$$

$$\langle\phi_1|\phi_1\rangle = 1, \langle\phi_2|\phi_2\rangle = 1, \langle\phi_1|\phi_2\rangle = 0, \langle\phi_2|\phi_1\rangle = 0$$

$$\langle\psi_1|\psi_1\rangle = 1 \Rightarrow |a_1|^2 + |a_2|^2 = 1$$

$$\langle\psi_2|\psi_2\rangle = 1 \Rightarrow b_1^*b_1\langle\phi_1|\phi_1\rangle + b_2^*b_2\langle\phi_2|\phi_2\rangle + b_1^*b_2\langle\phi_1|\phi_2\rangle + b_2^*b_1\langle\phi_2|\phi_1\rangle = 1$$

$$\text{if } \langle\phi_i|\phi_j\rangle = \delta_{i,j}, \langle\phi_1|\phi_1\rangle = 1, \langle\phi_2|\phi_2\rangle = 1, \langle\phi_1|\phi_2\rangle = 0, \langle\phi_2|\phi_1\rangle = 0$$

$$b_1^*b_1 + b_2^*b_2 = 1 \Rightarrow |b_1|^2 + |b_2|^2 = 1$$

$$\langle\psi_1|\psi_2\rangle = 0 \Rightarrow a_1^*b_1\langle\phi_1|\phi_1\rangle + a_2^*b_2\langle\phi_2|\phi_2\rangle + a_1^*b_2\langle\phi_1|\phi_2\rangle + a_2^*b_1\langle\phi_2|\phi_1\rangle = 0$$

$$\text{if } \langle\phi_i|\phi_j\rangle = \delta_{i,j}, \langle\phi_1|\phi_1\rangle = 1, \langle\phi_2|\phi_2\rangle = 1, \langle\phi_1|\phi_2\rangle = 0, \langle\phi_2|\phi_1\rangle = 0$$

$$\langle\psi_1|\psi_2\rangle = 0 \Rightarrow a_1^*b_1 + a_2^*b_2 = 0$$

$$\text{Similarly, } \langle\psi_2|\psi_1\rangle = 0 \Rightarrow b_1^*a_1 + b_2^*a_2 = 0$$

Example: The vector in three-dimensional space is defined as

$$|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \quad |\phi_3\rangle = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}. \text{ Prove that they are linearly independent}$$

For $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle$ linearly independent $c_1|\phi_1\rangle + c_2|\phi_2\rangle + c_3|\phi_3\rangle = 0$

$$c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 0 \text{ solving the equation we get } c_1 = 0, c_2 + c_3 = 0, c_2 - c_3 = 0$$

Which has unique solution $c_1 = c_2 = c_3 = 0$ so they are linearly independent

Example: Prove that $\phi_1 = x, \phi_2 = x^2, \phi_3 = x^3$ are linearly independent

$$c_1\phi_1 + c_2\phi_2 + c_3\phi_3 = 0 \Rightarrow c_1x + c_2x^2 + c_3x^3 = 0$$

equating coefficient, we get $c_1 = c_2 = c_3 = 0$ which ensure that x, x^2, x^3 are linearly independent

Example: Prove that $\phi_1 = \exp -x^2, \phi_2 = x \exp(-x^2)$

$$c_1\phi_1 + c_2\phi_2 = 0 \Rightarrow \exp(-\alpha x^2)[c_1x + c_2] = 0 \text{ hence } \exp(-\alpha x^2) \neq 0$$

$[c_1x + c_2] = 0 \Rightarrow c_1 = c_2 = 0$ which ensure $\phi_1 = \exp -x^2, \phi_2 = x \exp(-x^2)$ are linearly independent.

Example: (a) Prove that vector $|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|\phi_2\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ are linearly independent but not orthogonal

(b) Prove that vector $|\phi_1\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $|\phi_2\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ are orthogonal. Also check their linearly independency.

Solution: (a) $c_1|\phi_1\rangle + c_2|\phi_2\rangle = 0 \Rightarrow c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$

$\Rightarrow c_1 + c_2 = 0$ and $c_2 = 0$ which ensure $c_1 = 0, c_2 = 0$ that means $|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|\phi_2\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ are linearly independent.

Now we check orthogonal relation $\langle \phi_1 | \phi_2 \rangle = 0$

$$\Rightarrow \langle \phi_1 | \phi_2 \rangle = (1 \ 0) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \neq 0 \text{ which means } |\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |\phi_2\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ are not orthogonal}$$

So it is not necessary that linear independent vectors are orthogonal.

(b) If $|\phi_1\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $|\phi_2\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ we check orthogonal relation $\langle\phi_1|\phi_2\rangle = 0$

$$\Rightarrow \langle\phi_1|\phi_2\rangle = (1 \ 1) \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \times 1 + 1 \times (-1) = 0, \text{ which means } |\phi_1\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } |\phi_2\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ are}$$

orthogonal.

So, if vectors are orthogonal then they must be linearly independent.

Let's check there independency

$$c_1 |\phi_1\rangle + c_2 |\phi_2\rangle = 0 \Rightarrow c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0$$

$$\Rightarrow c_1 + c_2 = 0 \text{ and } c_1 - c_2 = 0 \text{ which ensure } c_1 = 0, c_2 = 0 \text{ that means } |\phi_1\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } |\phi_2\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

are linearly independent.

Chapter 1

Tools of Quantum Mechanics

4. Operators and Matrices

An operator $\hat{A}$ is the mathematical rule that when applied to a ket $|\phi\rangle$ will transformed into another $|\psi\rangle$ of the same space and when it acts on a any bra $\langle\chi|$ it transforms it into another bra $\langle\phi|$. that means $\hat{A}|\phi\rangle = |\psi\rangle$ and $\langle\chi|\hat{A} = \langle\phi|$

Similar definition is also applied on function as $\hat{A}\phi(r) = \psi(r)$

Basic example of operators Example of Operator:

Identity operator $I|\psi\rangle = |\psi\rangle$

Parity operator $\pi|\psi(r)\rangle = |\psi(-r)\rangle$

Gradient operator $\nabla\psi(r)$

Linear momentum operator $P(\psi) = -i\hbar\nabla\psi(\vec{r})$

Linear Operator: $\hat{A}$ is linear operator if

$$\hat{A}(|\psi\rangle + |\phi\rangle) = \hat{A}|\psi\rangle + \hat{A}|\phi\rangle$$

$$\hat{A}(c|\phi\rangle) = c\hat{A}|\phi\rangle$$

$$(\hat{A} + \hat{B})|\psi\rangle = \hat{A}|\psi\rangle + \hat{B}|\psi\rangle$$

$$\hat{A}(a_1|\psi_1\rangle + a_2|\psi_2\rangle) = a_1\hat{A}|\psi_1\rangle + a_2\hat{A}|\psi_2\rangle$$

$$(\langle\psi_1|a_1 + \langle\psi_2|a_2)\hat{A} = \langle\psi_1|\hat{A}a_1 + \langle\psi_2|\hat{A}a_2$$

If two operators are $\hat{A}$ and $\hat{B}$ then $\hat{A}\hat{B}$ is product of operator. Product of two linear operator $\hat{A}$ and $\hat{B}$ written $\hat{A}\hat{B}$ which is defined $(\hat{A}\hat{B})|\psi\rangle = \hat{A}(\hat{B}|\psi\rangle)$

In general, $\hat{A}\hat{B} \neq \hat{B}\hat{A}$

Matrix representation of operator Any operator $\hat{A}$ can be represented in square matrix of order $N \times N$ in a basis of $|\phi_1\rangle, |\phi_2\rangle, \dots, |\phi_i\rangle, \dots, |\phi_j\rangle, \dots, |\phi_N\rangle$ with matrix element $A_{i,j} = \langle\phi_i|\hat{A}|\phi_j\rangle$

Hermitian Adjoint of Operator

If operator $\hat{A}$ is defined as $\hat{A}|\phi\rangle = |\psi\rangle$ then $\langle\phi|\hat{A}^\dagger = \langle\psi|$ where $\hat{A}^\dagger$ is identified as Hermitian conjugate of operator $\hat{A}$. Hermitian conjugate $\hat{A}^\dagger$ of matrix A can be find in two step.

Step I: Find transpose of $\hat{A}$ i.e. Convert row into column ie $\hat{A}^T$

Step II: Then take complex conjugate to each element of $\hat{A}^T$ i.e. $\hat{A}^\dagger = (\hat{A}^T)^*$.

Properties of Hermitian adjoint operator $\hat{A}^\dagger$

- $(\hat{A}^\dagger)^\dagger = \hat{A}$
- $(\hat{A}^n)^\dagger = (\hat{A}^\dagger)^n$
- $(a\hat{A})^\dagger = a^* \hat{A}^\dagger$
- $(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$
- $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$

Proof of $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$

Let us assume $\hat{A}\hat{B}|\phi\rangle = |\psi\rangle \Rightarrow \langle\phi|(\hat{A}\hat{B})^\dagger = \langle\psi|$

In this process $\hat{B}|\phi\rangle = |\chi\rangle \Rightarrow \langle\phi|\hat{B}^\dagger = \langle\chi|$

$\hat{A}\hat{B}|\phi\rangle = \hat{A}|\chi\rangle = |\psi\rangle \Rightarrow \langle\chi|\hat{A}^\dagger = \langle\psi| \Rightarrow \langle\phi|\hat{B}^\dagger \hat{A}^\dagger = \langle\psi|$ so we can conclude $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$

Eigen value of operator:

If A operator is defined such that $\hat{A}|\phi\rangle = \lambda|\phi\rangle$ then

λ is said to be eigen value and $|\phi\rangle$ is said to be eigen vector corresponding to operator.

$(\hat{A} - \lambda\hat{I})|\phi\rangle = 0, |\phi\rangle \neq 0$ so $|\hat{A} - \lambda\hat{I}| = 0$ identify as characteristic equation

The values of λ satisfying the characteristic equation are known as eigen value if λ is non repeated then it is non degenerate eigen value and if λ is repeated then it is degenerate eigen value.

If the $N \times N$ matrix $\rightarrow N$ value of λ

For the given value of $\lambda = \lambda_n$ if any column vector $|\phi_n\rangle$ will satisfy $\hat{A}|\phi_n\rangle = \lambda_n|\phi_n\rangle$ then $|\phi_n\rangle$ is identified as eigen vector corresponds is eigen value λ_n . In Quantum Mechanics one should always find orthonormal set of eigen vectors such that it can make complete basis. For non-degenerate eigen values one can make unique set of orthonormal eigen state but for degenerate eigen values there may be more than one set of orthonormal vectors.

Hermitian Operators

An operator is said to be Hermitian if Hermitian adjoint is same operators which means $\hat{A}^\dagger = \hat{A}$ i.e., Matrix element $\langle\phi_i|\hat{A}|\phi_j\rangle = (\langle\phi_j|\hat{A}|\phi_i\rangle)^*$ which can be also written as

$$\int_{-\infty}^{\infty} \phi(x) (\hat{A}\psi(x))^* dx = \int_{-\infty}^{\infty} \psi^*(x) \hat{A}\phi(x) dx$$

- The eigen values of Hermitian matrix is real

$$\hat{A}|\phi_n\rangle = \lambda_n|\phi_n\rangle \Rightarrow \langle\phi_n|\hat{A}^\dagger = \lambda_n^* \langle\phi_n|$$

Now $\langle\phi_n|\hat{A}|\phi_n\rangle = \lambda_n \langle\phi_n|\phi_n\rangle \Rightarrow \langle\phi_n|\hat{A}^\dagger|\phi_n\rangle = \lambda_n^* \langle\phi_n|\phi_n\rangle$ hence $\hat{A}$ is Hermitian then $\hat{A} = \hat{A}^\dagger$

then $\langle\phi_n|\hat{A}|\phi_n\rangle = \lambda_n \langle\phi_n|\phi_n\rangle \Rightarrow \langle\phi_n|\hat{A}|\phi_n\rangle = \lambda_n^* \langle\phi_n|\phi_n\rangle$ equating both side we will get

$\lambda_n = \lambda_n^*$ which means eigen values are real

- The eigen vectors corresponding to different eigen values are orthogonal.

$$\hat{A}|\phi_n\rangle = \lambda_n|\phi_n\rangle \Rightarrow \langle\phi_m|\hat{A}|\phi_n\rangle = \lambda_n \langle\phi_m|\phi_n\rangle$$

$$\hat{A}|\phi_m\rangle = \lambda_m|\phi_m\rangle \Rightarrow \langle\phi_m|\hat{A}^\dagger = \lambda_m^* \langle\phi_m| \text{ where } m \neq n \Rightarrow \langle\phi_m|\hat{A}^\dagger|\phi_n\rangle = \lambda_m^* \langle\phi_m|\phi_n\rangle \text{ hence } \hat{A} \text{ is}$$

Hermitian then $\hat{A} = \hat{A}^\dagger \Rightarrow \langle\phi_m|\hat{A}|\phi_n\rangle = \lambda_m^* \langle\phi_m|\phi_n\rangle$

$$\langle\phi_m|\hat{A}|\phi_n\rangle - \langle\phi_m|\hat{A}|\phi_n\rangle = (\lambda_n - \lambda_m^*) \langle\phi_m|\phi_n\rangle \Rightarrow (\lambda_n - \lambda_m^*) \langle\phi_m|\phi_n\rangle = 0 \quad m \neq n$$

so $(\lambda_n \neq \lambda_m^*)$ i.e. $\langle \phi_m | \phi_n \rangle = 0$ which means eigen vectors corresponds to different eigen values are orthogonal

Some important operator and its property

Position Operator ($\hat{X}$): Mathematically position operator is defined as $\hat{X}\psi(x) = x\psi(x)$

Momentum Operators ($\hat{P}$): It is Mathematically define as $\hat{P} = -i\hbar\nabla$ the eigen value of Momentum operator is momentum

Hamiltonian Operator ($\hat{H}$): It is defined as $\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(x, y, z)$ The eigen value of Hamiltonian operator is Energy.

Parity Operator ($\hat{\pi}$): A Parity Operator (also called **parity inversion**) is the flip in the sign of spatial coordinate In three dimensions, it can also refer to the simultaneous flip in the sign of all three spatial coordinates (a Point reflection).

Mathematically it is defined as $\hat{\pi}\psi(x, y, z) = \psi(-x, -y, -z)$.

If $\hat{\pi}\psi(x, y, z) = \psi(-x, -y, -z) = \psi(x, y, z)$ then wave function is said to be symmetric or even symmetry, so even parity have eigen value 1

If $\hat{\pi}\psi(x, y, z) = \psi(-x, -y, -z) = -\psi(x, y, z)$ then wave function is said to be Antisymmetric symmetric or odd symmetry, so even parity have eigen value -1

So eigen value of parity is either $+1$ or -1 .

Translation Operator $\hat{T}(a)$: The translation operator moves particles and fields by the amount a Therefore, if a particle is in an eigen state of the Position operator (i.e., $\psi(x)$ precisely located at the position x then after operating $\hat{T}(a)$ on it, the particle is at the position $x + a$. $\hat{T}\psi(x) = \psi(x + a)$. Momentum operator is generator of translation operator. It

is defined as $T = \exp\left(\frac{i\hat{P}_x a}{\hbar}\right)$, where $\hat{P}_x = \frac{\partial}{\partial x}$

$$T\psi(x) = \exp\left(\frac{i\hat{P}_x a}{\hbar}\right)\psi(x) \Rightarrow \exp\left(a \frac{\partial}{\partial x}\right)\psi(x) = \sum_{N=0}^{\infty} \frac{1}{N!} \left(a \frac{\partial}{\partial x}\right)^N \psi(x)$$

$$\hat{T}\psi(x) = \psi(x) + a \frac{\partial \psi}{\partial x} + \frac{1}{2} a^2 \frac{\partial^2 \psi}{\partial x^2} + \dots = \psi(x + a)$$

Projection operator and completeness relation

If we want to project any vector $|\psi\rangle$ in direction of $|\phi_i\rangle$ then we need projection operator which

is define as $\hat{p}_i = |\phi_i\rangle\langle\phi_i|$. In general, $\langle\phi_i|\phi_j\rangle = \delta_{i,j}$

$\hat{p}_i|\psi\rangle = |\phi_i\rangle\langle\phi_i|\psi\rangle = c_i|\phi_i\rangle$ where c_i is complex number.

hence $\langle\phi_i|\phi_i\rangle = 1$ $\hat{p}_i^2 = \hat{p}_i \cdot \hat{p}_i = |\phi_i\rangle\langle\phi_i| = \hat{p}_i \Rightarrow \hat{p}_i^2 = \hat{p}_i \Rightarrow \hat{p}_i = O$ null operator or $p_i = I$ identity operator . so eigen value of projection operator is 0 or 1

if we have $\langle\phi_i|\phi_j\rangle = \delta_{i,j}$ and $\sum_{i=1}^N p_i = I \Rightarrow \sum_{i=1}^N |\phi_i\rangle\langle\phi_i| = I$ is identify as completeness relation in

quantum mechanics. The completeness relation can be use to write any ket vector $|\psi\rangle$ in a

orthonormal basis of $|\phi_1\rangle, |\phi_2\rangle \dots |\phi_i\rangle \dots |\phi_j\rangle \dots |\phi_N\rangle$ as $|\psi\rangle = I|\psi\rangle = \sum_{i=1}^N |\phi_i\rangle\langle\phi_i|\psi\rangle = |\psi\rangle = \sum_{i=1}^N c_i |\phi_i\rangle$

where $\langle\phi_i|\psi\rangle = c_i$

In quantum mechanics orthonormal set of eigen values of Hermitian operators will make complete basis.

Example: If $\hat{A}$ is an operator $\hat{A} = \hat{D}_x + \hat{X}$, where $\hat{D}_x\psi(x) = \frac{d\psi}{dx}$ and $\hat{X}\psi(x) = x\psi(x)$ find equivalent of $\hat{A}^2$.

Solution: $\hat{A}^2 = (\hat{D}_x + \hat{X})^2 = (\hat{D}_x + \hat{X})(\hat{D}_x + \hat{X})$

$$A^2\psi(x) = \left(\frac{d}{dx} + \hat{X}\right)\left(\frac{d}{dx} + \hat{X}\right)\psi(x) = \frac{d^2\psi(x)}{dx^2} + \hat{X}\frac{d\psi(x)}{dx} + \psi(x) + \frac{\hat{X}d\psi(x)}{dx} + \hat{X}^2\psi(x)$$

$$A^2\psi(x) = \frac{d^2\psi}{dx^2} + 2x\frac{d\psi}{dx} + x^2\psi(x) + \psi(x)$$

$$\Rightarrow A^2 \equiv \frac{d^2}{dx^2} + 2x\frac{d}{dx} + x^2 + 1 \Rightarrow \hat{A}^2 = \hat{D}_x^2 + 2\hat{X}\hat{D}_x + \hat{X}^2 + I$$

Example: If $\hat{D}_x$ is defined as $\frac{\partial}{\partial x}$ and $\psi(x) = A \sin \frac{n\pi x}{a}$

(a) operate $\hat{D}_x$ and $\psi(x)$

(b) operate $\hat{D}_x^2$ on $\psi(x)$

(c) which one of the above given eigen value problem.

Solution: (a) $\hat{D}_x \psi(x) = \frac{\partial}{\partial x} A \sin \frac{n\pi x}{a} = A \frac{n\pi}{a} \cos \frac{n\pi x}{a}$

(b) $\hat{D}_x^2 \psi(x) = \frac{\partial^2}{\partial x^2} A \sin \frac{n\pi x}{a} = A \left(-\frac{n^2 \pi^2}{a^2} \right) \sin \frac{n\pi x}{a}$

$$\hat{D}_x^2 \psi(x) = -\frac{n^2 \pi^2}{a^2} A \sin \frac{n\pi x}{a}$$

(c) when $\hat{D}_x^2$ operate on $\hat{D}_x^2 \psi(x) = -\frac{n^2 \pi^2}{a^2} \psi(x)$

So operation of $\hat{D}_x^2(x)$ on $\psi(x) = A \sin \frac{n\pi x}{a}$ give eigen value problem with eigen value $-\frac{n^2 \pi^2}{a^2}$

Example: The momentum operator in one dimension is defined as $\hat{P}_x = -i\hbar \frac{\partial}{\partial x}$.

(a) If p is eigen value of momentum operator find corresponding the eigen function.

(b) If particle is confine between $-\frac{l}{2} \leq x \leq \frac{l}{2}$ then write down Normalized eigen state.

Solution: (a) Let us assume $\psi(x)$ is eigen function of momentum operator with eigen value p .

$\hat{P}_x \psi(x) = p\psi \Rightarrow -i\hbar \frac{\partial \psi}{\partial x} = p\psi \Rightarrow -i\hbar \int \frac{\partial \psi}{\psi} = p \int dx \Rightarrow \psi = A \exp\left(\frac{ipx}{\hbar}\right)$ where A is normalized constant.

(b) $\int_{-\frac{l}{2}}^{\frac{l}{2}} \psi^*(x) \psi(x) dx \Rightarrow |A|^2 \int_{-\frac{l}{2}}^{\frac{l}{2}} \exp \frac{-ipx}{\hbar} \cdot \exp \frac{ipx}{\hbar} dx = 1 \Rightarrow A = \frac{1}{\sqrt{l}}$

The normalized wave function is given as $\psi(x) = \begin{cases} \frac{1}{\sqrt{l}} \exp \frac{ipx}{\hbar}, & -\frac{l}{2} \leq x \leq \frac{l}{2} \\ 0, & \text{otherwise} \end{cases}$

Example: Two operators are given as $\hat{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & -3i \\ 0 & 3i & 5 \end{pmatrix}$ and $\hat{B} = \begin{pmatrix} 0 & -i & 3i \\ -i & 0 & i \\ 3i & i & 0 \end{pmatrix}$, which of following have real eigen values.

Solution: $\hat{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & -3i \\ 0 & 3i & 5 \end{pmatrix}$ $\hat{A}^\dagger = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & -3i \\ 0 & 3i & 5 \end{pmatrix}$

$\hat{A}^\dagger = \hat{A}$ so A is Hermitian so it has real eigen value

$$\hat{B} = \begin{pmatrix} 0 & -i & -3i \\ -i & 0 & i \\ 3i & i & 0 \end{pmatrix} \quad \hat{B}^\dagger = \begin{pmatrix} 0 & -i & -3i \\ i & 0 & -i \\ -3i & -i & 0 \end{pmatrix}$$

$\hat{B}^\dagger = -\hat{B}$ So, it is not Hermitian rather it is Anti-Hermitian so its eigen values are not real.

Example: If $\hat{S}$ operator is defined as $\hat{S}|u_1\rangle = |u_3\rangle$, $\hat{S}|u_2\rangle = |u_2\rangle$, $\hat{S}|u_3\rangle = |u_1\rangle$. It is given $\langle u_i | u_j \rangle = \delta_{ij}$ represent operator $\hat{S}$ in basis of $|u_1\rangle, |u_2\rangle$ and $|u_3\rangle$.

Solution: The Matrix $\hat{S} = \begin{pmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{pmatrix}$

Where matrix element $S_{ij} = \langle u_i | \hat{S} | u_j \rangle$

$$S_{11} = \langle u_1 | \hat{S} | u_1 \rangle = \langle u_1 | u_3 \rangle = 0$$

$$S_{12} = \langle u_1 | \hat{S} | u_2 \rangle = \langle u_1 | u_2 \rangle = 0$$

$$S_{13} = \langle u_1 | \hat{S} | u_3 \rangle = \langle u_1 | u_3 \rangle = 1$$

$$S_{21} = \langle u_2 | \hat{S} | u_1 \rangle = \langle u_2 | u_3 \rangle = 0$$

$$S_{22} = \langle u_2 | \hat{S} | u_2 \rangle = \langle u_2 | u_2 \rangle = 1$$

$$S_{23} = \langle u_2 | \hat{S} | u_3 \rangle = \langle u_2 | u_1 \rangle = 0$$

$$S_{31} = \langle u_3 | \hat{S} | u_1 \rangle = \langle u_3 | u_3 \rangle = 1$$

$$S_{32} = \langle u_3 | \hat{S} | u_2 \rangle = \langle u_3 | u_2 \rangle = 0$$

$$S_{33} = \langle u_3 | \hat{S} | u_3 \rangle = \langle u_3 | u_1 \rangle = 0$$

$$\hat{S} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Example: If $\hat{A}$ operator is given by $\hat{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(a) Find eigen value and eigen vector of $\hat{A}$.

(b) Normalized there eigen vector.

(c) Prove both eigen vector are orthogonal.

Solution: (a) $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ for eigen value

$$|A - \lambda I| = 0$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 0 \Rightarrow \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda = 1, \text{ and } \lambda = -1,$$

The eigen vector corresponding to $\lambda = 1$,

$$A|u_1\rangle = \lambda|u_1\rangle$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow a = b$$

So eigen vector corresponds to $\lambda = 1$, $|u_1\rangle = \begin{pmatrix} a \\ a \end{pmatrix}$

eigen vector corresponds to $\lambda = -1$

$$A|u_2\rangle = \lambda|u_2\rangle$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = -1 \begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow a = -b \Rightarrow |u_2\rangle = \begin{pmatrix} a \\ -a \end{pmatrix}$$

(b) For normalised eigen vector.

$$\langle u_1 | u_1 \rangle = 1 \quad \langle u_2 | u_2 \rangle = 1$$

$$|u_1\rangle = \begin{pmatrix} a \\ a \end{pmatrix} \quad \langle u_1| = (a \quad a)$$

$$\langle u_1 | u_1 \rangle = a^2 + a^2 = 1 \Rightarrow a = \frac{1}{\sqrt{2}} \quad |u_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Similarly, $|u_2\rangle = \begin{pmatrix} a \\ -a \end{pmatrix}$ and $\langle u_2| = (a, -a)$

$$\langle u_2 | u_2 \rangle = (a, -a) \begin{pmatrix} a \\ -a \end{pmatrix} = 1 \Rightarrow a^2 + a^2 = 1 \Rightarrow a = \frac{1}{\sqrt{2}} \Rightarrow |u_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

(c) For orthogonality $\langle u_1 | u_2 \rangle = \langle u_2 | u_1 \rangle = 0$

$$\frac{1}{\sqrt{2}}(1 \quad 1) \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0 \quad \frac{1}{\sqrt{2}}(1 \quad -1) \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$$

Example: (a) Find the eigen value and orthonormal eigen vectors of operator $\hat{A} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

(b) Prove that eigen vectors of $\hat{A}$ are satisfying completeness relation

$$|\hat{A} - \lambda \hat{I}| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & -\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} = 0 \Rightarrow (2-\lambda)(\lambda^2 - 1) = 0$$

Solution: Eigen values are $\lambda_1 = 2, \lambda_2 = 1, \lambda_3 = -1$ hence they are non-repeated they are nondegenerate eigen values

Now let us find Eigen vector for eigen value $\lambda_1 = 2$, which is $|\phi_1\rangle = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix}$

$$\hat{A}|\phi_1\rangle = \lambda_1|\phi_1\rangle \Rightarrow \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} = 2 \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix}$$

$2a_1 = 2a_1, c_1 = 2b_1, b_1 = 2c_1$ the solutions are $b_1 = c_1 = 0$ and a_1 can be any arbitrary value so

$|\phi_1\rangle = \begin{pmatrix} a_1 \\ 0 \\ 0 \end{pmatrix}$ we can find value of a_1 with normalization condition $\langle\phi_1|\phi_1\rangle = 1$ Normalization to get

one specific vector

$$\begin{pmatrix} a_1^* & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ 0 \\ 0 \end{pmatrix} = 1 \Rightarrow |a_1|^2 = 1 \Rightarrow a_1 = 1$$

Now let us find Eigen vector for eigen value $\lambda_2 = 1$, which is $|\phi_2\rangle = \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix}$

$$\hat{A}|\phi_2\rangle = \lambda_2|\phi_2\rangle \Rightarrow \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix} = 1 \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix} = \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix} \Rightarrow 2a_2 = a_2 \Rightarrow a_2 = 0 \quad c_2 = b_2, b_2 = c_2 \Rightarrow b_2 = c_2$$

$$|\phi_2\rangle = \begin{pmatrix} 0 \\ b_2 \\ b_2 \end{pmatrix} \text{ the value of } b_2 \text{ can be found with normalization condition}$$

$$\langle \phi_2 | \phi_2 \rangle = \begin{pmatrix} 0 & b_2 & b_2 \end{pmatrix} \begin{pmatrix} 0 \\ b_2 \\ b_2 \end{pmatrix} = 2|b_2|^2 = 1 \Rightarrow b_2 = \frac{1}{\sqrt{2}} \quad |\phi_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Now we will find eigen vector for eigen value $\lambda_3 = -1$ which is $|\phi_3\rangle = \begin{pmatrix} a_3 \\ b_3 \\ c_3 \end{pmatrix}$

$$\hat{A}|\phi_3\rangle = \lambda_3 |\phi_3\rangle \Rightarrow \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_3 \\ b_3 \\ c_3 \end{pmatrix} = -1 \begin{pmatrix} a_3 \\ b_3 \\ c_3 \end{pmatrix}$$

$$2a_3 = -a_3 \Rightarrow a_3 = 0, c_3 = -b_3, b_3 = -c_3 \quad |\phi_3\rangle = \begin{pmatrix} 0 \\ b_3 \\ -b_3 \end{pmatrix}$$

The value of b_3 can be found from normalization condition

$$\langle \phi_3 | \phi_3 \rangle = \begin{pmatrix} 0 & b_3^* & -b_3^* \end{pmatrix} \begin{pmatrix} 0 \\ b_3 \\ -b_3 \end{pmatrix} = 1 \Rightarrow b_3 = 1 \quad |\phi_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$|\phi_1\rangle, |\phi_2\rangle$ and $|\phi_3\rangle$ are orthogonal set.

$$|\phi_1\rangle\langle\phi_1| + |\phi_2\rangle\langle\phi_2| + |\phi_3\rangle\langle\phi_3| = I$$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

If Eigen value is non-degenerate, then one can find unique set of orthonormal basis vector.

Example: Find the eigen values of operator $\hat{A} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix}$

Find two set of orthonormal eigen vectors.

Solution: $[\hat{A} - \lambda I] = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & -\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} = 0 \Rightarrow (1-\lambda)(\lambda^2 - 1) = 0 \Rightarrow \lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 1$

Hence $\lambda_2 = \lambda_3 = 1$ so eigen value 1 is doubly degenerate.

Eigen vectors for $\lambda_1 = -1$ is $|\phi_1\rangle = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix}$

$$\hat{A}|\phi_1\rangle = \lambda_1|\phi_1\rangle \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} = -1 \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} \Rightarrow a_1 = -a_1 \Rightarrow a_1 = 0, c_1 = -b_1 \text{ or } b_1 = -c_1$$

$$|\phi_1\rangle = \begin{pmatrix} 0 \\ b_1 \\ -b_1 \end{pmatrix} \text{ using normalization condition } \langle \phi_1 | \phi_1 \rangle = 1 \Rightarrow b_1 = \frac{1}{\sqrt{2}} \text{ so } |\phi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

Hence $\lambda_1 = -1$ is non degenerate Eigen value it has unique normalized vector.

Now we will find eigen vector for eigen value $\lambda = 1$ which is doubly degenerate so there may be infinite set of normalized eigen vectors but it is not necessary that they are orthogonal. so for $\lambda_2 = 1$ we need to choose two normal vectors which are orthogonal in nature

$$\hat{A}|\phi_2\rangle = \lambda_2|\phi_2\rangle \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix} = 1 \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix} \Rightarrow a_2 = a_2, c_2 = b_2, b_2 = c_2$$

$$|\phi_2\rangle = \begin{pmatrix} a_2 \\ b_2 \\ b_2 \end{pmatrix} \text{ where } a_2 \text{ and } b_2 \text{ has arbitrary number so there is possibility of infinite normalized}$$

vector.

Similarly, for $\lambda_3 = 1$

$$\hat{A}|\phi_3\rangle = \lambda_3|\phi_3\rangle \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_3 \\ b_3 \\ c_3 \end{pmatrix} = 1 \begin{pmatrix} a_3 \\ b_3 \\ c_3 \end{pmatrix} \Rightarrow a_3 = a_3, c_3 = b_3, b_3 = c_3$$

$|\phi_3\rangle = \begin{pmatrix} a_3 \\ b_3 \\ b_3 \end{pmatrix}$ where a_3 and b_3 has arbitrary number, so there is possibility of infinite normalized

vector

We can easily conclude $|\phi_2\rangle$ and $|\phi_3\rangle$ are vectors in same plane with is surely orthogonal to $|\phi_1\rangle$.

Now we need to choose two set of orthonormal vectors which are in same plane where $|\phi_1\rangle$ is uniquely defined for all set.

Set I: We can choose $a_2 = 1$, $|\phi_2\rangle = \begin{pmatrix} 1 \\ b_2 \\ b_2 \end{pmatrix}$

from normalization condition we can find $b_2 = 0$ so $|\phi_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Now we can not choose $|\phi_3\rangle = \begin{pmatrix} a_3 \\ b_3 \\ b_3 \end{pmatrix}$ it must be orthogonal to $|\phi_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$\langle \phi_2 | \phi_3 \rangle = 0 \Rightarrow (1 \ 0 \ 0) \begin{pmatrix} a_2 \\ b_2 \\ b_2 \end{pmatrix} = 0 \Rightarrow a_2 = 0$ so $|\phi_3\rangle$ must be in form of $|\phi_3\rangle = \begin{pmatrix} 0 \\ b_3 \\ b_3 \end{pmatrix}$

from normalization condition we can get $|\phi_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

So, first set of orthonormal vectors are corresponds to eigen value $-1, 1, 1$ is

$|\phi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$, $|\phi_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $|\phi_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$. Again we can easily check these vectors make complete basis.

Set II: Again eigen value of nondegenerate eigen vector is $|\phi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ which is uniquely defined but we can choose another set of orthonormal eigen vectors for degenerate eigen value

$$\lambda_2 = \lambda_3 = 1 \quad |\phi_2\rangle = \begin{pmatrix} a_2 \\ b_2 \\ b_2 \end{pmatrix} \text{ and } |\phi_3\rangle = \begin{pmatrix} a_3 \\ b_3 \\ b_3 \end{pmatrix}$$

We can choose arbitrary $|\phi_2\rangle = \begin{pmatrix} a_2 \\ b_2 \\ b_2 \end{pmatrix}$ by putting $a_2 = \frac{1}{2}$, $|\phi_2\rangle = \begin{pmatrix} 1/2 \\ b_2 \\ b_2 \end{pmatrix}$

From normalization condition $\langle \phi_2 | \phi_2 \rangle = \frac{1}{4} + 2b_2^2 = 1$ $b_2 = \sqrt{\frac{3}{8}}$ so $|\phi_2\rangle = \begin{pmatrix} 1/2 \\ \sqrt{3/8} \\ \sqrt{3/8} \end{pmatrix}$

Now $|\phi_3\rangle = \begin{pmatrix} a_3 \\ b_3 \\ b_3 \end{pmatrix}$ cannot be chosen arbitrary it must be orthogonal to $|\phi_2\rangle = \begin{pmatrix} 1/2 \\ \sqrt{3/8} \\ \sqrt{3/8} \end{pmatrix}$

$$\langle \phi_2 | \phi_3 \rangle = 0 \Rightarrow \left(\frac{1}{2} \sqrt{\frac{3}{8}} \sqrt{\frac{3}{8}} \right) \begin{pmatrix} a_3 \\ b_3 \\ b_3 \end{pmatrix} = 0 \Rightarrow \frac{1}{2} a_3 + 2 \sqrt{\frac{3}{8}} b_3 = 0 \Rightarrow a_3 = -b_3 4 \sqrt{\frac{3}{8}}$$

$|\phi_3\rangle = \begin{pmatrix} -b_3 4 \sqrt{\frac{3}{8}} \\ b_3 \\ b_3 \end{pmatrix}$ the value of b_3 can be found with normalization condition

$$\langle \phi_3 | \phi_3 \rangle = 1 \Rightarrow 16 \times \frac{3}{8} b_3^2 + 2b_3^2 = 1 \Rightarrow 8b_3^2 = 1 \Rightarrow b_3 = \sqrt{\frac{1}{8}} \Rightarrow |\phi_3\rangle = \begin{pmatrix} -\sqrt{3/2} \\ 1/\sqrt{8} \\ 1/\sqrt{8} \end{pmatrix}$$

So second set of orthonormal vectors corresponding to eigen values are $\lambda_1 = -1, \lambda_2 = 1$ and $\lambda_3 = 1$

are $|\phi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ $|\phi_2\rangle = \begin{pmatrix} 1/2 \\ \sqrt{3/8} \\ \sqrt{3/8} \end{pmatrix}$ $|\phi_3\rangle = \begin{pmatrix} -\sqrt{3/2} \\ 1/\sqrt{8} \\ 1/\sqrt{8} \end{pmatrix}$ which will also make complete basis.

Example: Operate Parity operator π on functions $\psi_1 = \sin x$, $\psi_2 = \cos x$ and $\psi_3 = \sin x + \cos x$, also discuss parity symmetry of each function.

Solution: $\pi\psi_1 = \pi \sin x = \sin(-x) = -\sin x = -\psi_1$ which is antisymmetric so eigen value is -1 .

$\pi\psi_2 = \pi \cos x = \cos(-x) = \cos x = \psi_2$ which is symmetric so eigen value is 1 .

$\pi\psi_3 = \pi \sin x + \pi \cos x = \sin(-x) + \cos(-x) = -\sin x + \cos x \neq \psi_3$ so ψ_3 is not eigen function of parity operator.

Chapter 1

Tools of Quantum Mechanics

5. Commutator in Quantum Mechanics

Commutator: if $\hat{A}$ and $\hat{B}$ are two operators then the commutator $[\hat{A}, \hat{B}]$ is defined as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

Properties of Commutator:

- Antisymmetric $[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$
- Linearity $[\hat{A}, \hat{B} + \hat{C} + \hat{D}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}] + [\hat{A}, \hat{D}]$
- Distributivity: $[\hat{A}\hat{B}, \hat{C}] = [\hat{A}, \hat{C}]\hat{B} + \hat{A}[\hat{B}, \hat{C}]$
 $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$
- Jacobi Identity $[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$
- Hermitian conjugate to commutator $([\hat{A}, \hat{B}])^\dagger = [\hat{B}^\dagger, \hat{A}^\dagger]$
- $[\hat{A}, a] = 0$

$$\bullet \quad [\hat{A}, \hat{B}^n] = \sum_{j=0}^{n-1} \hat{B}^j [\hat{A}, \hat{B}] \hat{B}^{n-j-1}$$

$$[\hat{A}^n, \hat{B}] = \sum_{j=0}^{n-1} \hat{A}^{n-j-1} [\hat{A}, \hat{B}] \hat{A}^j$$

Commuting operators and Simultaneous Eigenfunctions

Consider two operators $\hat{A}$ and $\hat{B}$ which represent observables and are therefore Hermitian. If there is a $|\psi\rangle$ such that $\hat{A}|\psi\rangle = a|\psi\rangle$ and $\hat{B}|\psi\rangle = b|\psi\rangle$, then $|\psi\rangle$ is a simultaneous eigenfunction of $\hat{A}$ and $\hat{B}$, belonging to eigenvalues a and b , respectively.

Hence $\hat{B}\hat{A}|\psi\rangle = \hat{B}(a|\psi\rangle) = a\hat{B}|\psi\rangle = ab|\psi\rangle$ and $\hat{A}\hat{B}|\psi\rangle = \hat{A}(b|\psi\rangle) = b\hat{A}|\psi\rangle = ba|\psi\rangle$

$\Rightarrow (\hat{A}\hat{B} - \hat{B}\hat{A})|\psi\rangle = 0$ Thus $|\psi\rangle$ is also an eigenfunction of $[\hat{A}, \hat{B}]$, belonging to the eigenvalue zero.

If $\hat{A}$ and $\hat{B}$ commute, then $[\hat{A}, \hat{B}]|\psi\rangle = 0$ holds for any $|\psi\rangle$.

Now we can show that the eigenfunctions of two commuting operators can always be constructed in such a way that they are simultaneous eigenfunctions.

Proof: Suppose $|\psi\rangle$ is an eigenfunction of $\hat{A}$, so

$$\hat{A}|\psi\rangle = a|\psi\rangle \text{ and suppose } \hat{A} \text{ and } \hat{B} \text{ commute.}$$

$$\text{Then } \hat{B}\hat{A}|\psi\rangle = \hat{A}\hat{B}|\psi\rangle = \hat{B}a|\psi\rangle = a\hat{B}|\psi\rangle$$

$$\text{or } \hat{A}(\hat{B}|\psi\rangle) = a(\hat{B}|\psi\rangle)$$

Thus, if we assume $\hat{B}|\psi\rangle \neq 0$, the function $\hat{B}|\psi\rangle$ is an eigenfunction of $\hat{A}$ belonging to the same eigenvalue, a . Two cases to consider:

(i) Nondegenerate case: If there is only one state $|\psi\rangle$ corresponding to the eigenvalue a , then

$$\hat{B}|\psi\rangle \text{ can differ from } |\psi\rangle \text{ only by a constant factor, or } \hat{B}|\psi\rangle = b|\psi\rangle.$$

Hence, in this case, $[\hat{A}, \hat{B}]|\psi\rangle = 0$ is both a necessary and sufficient condition for $|\psi\rangle$ to be a simultaneous eigenfunction of $\hat{A}$ and $\hat{B}$.

(ii) Degenerate Case: Suppose, for example, that there are two states corresponding to the eigenvalue a , $|\psi_1\rangle$ and $|\psi_2\rangle$. These must be linearly independent and may for convenience be assumed to be orthonormal. Any linear combination $|\psi\rangle = c_1|\psi_1\rangle + c_2|\psi_2\rangle$ is also an

eigenfunction of $\hat{A}$ but need not be an eigenfunction of $\hat{B}$. However, we can try to determine a particular linear combination(s), which will be an eigenfunction of $\hat{B}$, such that

$$\hat{B}(c_1|\psi_1\rangle + c_2|\psi_2\rangle) = b(c_1|\psi_1\rangle + c_2|\psi_2\rangle)$$

Taking scalar products with $|\psi_1\rangle$ and $|\psi_2\rangle$, respectively, we obtain

$$\hat{B}_{11} = \langle\psi_1|\hat{B}|\psi_1\rangle, \hat{B}_{22} = \langle\psi_2|\hat{B}|\psi_2\rangle, \hat{B}_{12} = \langle\psi_1|\hat{B}|\psi_2\rangle, \hat{B}_{21} = \langle\psi_2|\hat{B}|\psi_1\rangle$$

$$c_1\hat{B}_{11} + c_2\hat{B}_{12} = bc_1 + c_2 \cdot 0 \Rightarrow c_1\hat{B}_{11} + c_2\hat{B}_{12} = bc_1$$

$$c_1\hat{B}_{21} + c_2\hat{B}_{22} = c_1 \cdot 0 + bc_2 \Rightarrow c_1\hat{B}_{21} + c_2\hat{B}_{22} = bc_2$$

$$c_1(\hat{B}_{11} - b) + c_2\hat{B}_{12} = 0 \text{ with, etc.}$$

$$c_1\hat{B}_{21} + c_2(\hat{B}_{22} - b) = 0$$

Nontrivial solution exists if
$$\begin{vmatrix} \hat{B}_{11} - b & \hat{B}_{12} \\ \hat{B}_{21} & \hat{B}_{22} - b \end{vmatrix} = 0$$

If the two roots b_1 and b_2 of the secular equation are distinct, the corresponding eigenfunctions of $\hat{B}$ are linearly independent. Hence the degeneracy has been resolved via the commuting operator $\hat{B}$.

If the roots b_1 and b_2 are the same, then if $\hat{B}$ is Hermitian, $\hat{B}_{11} = \hat{B}_{22}$ and $\hat{B}_{12} = \hat{B}_{21}^* = 0$ are necessary and sufficient conditions. Accordingly any c_1 and c_2 can be used; i.e., $|\psi_1\rangle$ and $|\psi_2\rangle$ are simultaneous, degenerate eigenfunctions of $\hat{B}$ as well as $\hat{A}$.

The analysis can be generalized immediately to cases of N -fold degeneracy. If the secular equation in b has N distinct roots, the degeneracy is completely resolved.

If the roots are not all distinct, it may be possible to find a third operator $\hat{C}$ which commutes with $\hat{A}$ and $\hat{B}$ and which will further resolve the remaining degeneracy, and do on. Ultimately, an unambiguous description of the system can be obtained in terms of a complete set of commuting observables, the eigenvalues of which provide a unique, nondegenerate specification of each state.

Set of commuting observables

- If two operators A and B commute and if $|\psi\rangle$ is eigen vector of A , then $B|\psi\rangle$ is also an eigen vector of A , with the same eigen value.

- If two operators A and B commute and if $|\psi_1\rangle$ and $|\psi_2\rangle$ are two eigen vector of A with different eigen values then Matrix element $\langle\psi_1|B|\psi_2\rangle$ is zero.
- If two operators A and B commute one can construct an orthonormal basis of state space with eigen vectors common to A and B .

Functions of Operator

Function of linear operator $F(\hat{A})$ can be expand as polynomial of $\hat{A}$ as $F(\hat{A}) = \sum_n a_n \hat{A}^n$

For example, $\exp(a\hat{A}) = \sum_n \frac{a^n}{n!} \hat{A}^n$

If two operator $F(\hat{A})$ and $G(\hat{A})$ are function of operator $\hat{A}$ then commutator

$$[F(\hat{A}), G(\hat{A})] = 0$$

If commutator $[\hat{A}, \hat{B}] = 0$ then $[B, F(\hat{A})] = 0$

$$[F(\hat{A})]^\dagger = F^*(\hat{A}^\dagger) \text{ for example, } [\exp(ia\hat{A})]^\dagger = \exp(-ia^* \hat{A}^\dagger)$$

$$\exp(\hat{A}) \cdot \exp(\hat{B}) = \exp(\hat{A} + \hat{B}) \cdot \exp\left[\frac{[\hat{A}, \hat{B}]}{2}\right]$$

$$\exp(\hat{A}) \hat{B} \exp(-\hat{A}) = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!} [\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots$$

Example: If momentum operator $\hat{P}_x$ is defined as $-i\hbar \frac{\partial}{\partial x}$ and position operator $\hat{X}$ is defined as

$$\hat{X}\psi(x) = x\psi(x)$$

(a) Find the value of commutator $[\hat{X}, \hat{P}_x]$

(b) Using the property of commutator find the value of $[\hat{X}^2, \hat{P}_x]$ and $[\hat{X}, \hat{P}_x^2]$

(c) Using the property find the value of $[\hat{X}^2, \hat{P}_x^2]$

Solution: (a): $[\hat{X}, \hat{P}_x] = (\hat{X} \hat{P}_x - \hat{P}_x \hat{X})$

Operator both side with ψ

$$\hat{X}\hat{P}_x\psi(x) - \hat{P}_x\hat{X}\psi(x)$$

$$\hat{X}(-i\hbar)\frac{\partial\psi(x)}{\partial x} - \hat{P}_x x\psi(x)$$

$$\hat{X}(-i\hbar)\frac{\partial\psi(x)}{\partial x} + i\hbar\frac{\partial}{\partial x}x\psi(x)$$

$$x(-i\hbar)\frac{\partial\psi(x)}{\partial x} + x i\hbar\frac{\partial\psi(x)}{\partial x} + i\hbar\frac{\partial x}{\partial x}\psi(x) = i\hbar\psi(x)$$

$$[\hat{X}, \hat{P}_x] = i\hbar\psi(x) \quad [\hat{X}, \hat{P}_x] = i\hbar$$

$$(b) \quad [\hat{X}^2, \hat{P}_x] = [\hat{X} \cdot \hat{X}, \hat{P}_x] = \hat{X}[\hat{X}, \hat{P}_x] + [\hat{X}, \hat{P}_x]\hat{X} = \hat{X}i\hbar + i\hbar\hat{X} = 2i\hbar\hat{X}$$

$$[\hat{X}, \hat{P}_x^2] = [\hat{X}, \hat{P}_x \cdot \hat{P}_x]$$

$$\hat{P}_x[\hat{X}, \hat{P}_x] + [\hat{X}, \hat{P}_x]\hat{P}_x = \hat{P}_xi\hbar + i\hbar\hat{P}_x \quad [\hat{X}, \hat{P}_x^2] = 2i\hbar\hat{P}_x$$

$$(c) \quad [\hat{X}^2, \hat{P}_x^2] = \hat{X}[\hat{X}, \hat{P}_x^2] + [\hat{X}, \hat{P}_x^2]\hat{X} \Rightarrow \hat{X}2i\hbar\hat{P}_x + 2i\hbar\hat{P}_x\hat{X} = 2i\hbar(\hat{X}\hat{P}_x + \hat{P}_x\hat{X})$$

Example: If $\hat{A} = a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\hat{B} = b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(a) Find the value of $[\hat{A}, \hat{B}]$

(b) Write down eigen vector of $\hat{B}$ in the basis of eigen vector of A .

Solution: (a) $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
 $= ab \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - ba \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} [\hat{A}, \hat{B}]$ so A and B commute.

(b) Eigen vector of A is $|a_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ eigen value $\lambda_1 = a$

$|a_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ eigen value $\lambda_2 = a$

Eigen vector of B is $|b_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ eigen value $\lambda_1 = b$

$|b_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ eigen value $\lambda_2 = -b$

$$|b_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} |a_1\rangle + \frac{1}{\sqrt{2}} |a_2\rangle$$

$$|b_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad |b_2\rangle = \frac{1}{\sqrt{2}} |a_1\rangle - \frac{1}{\sqrt{2}} |a_2\rangle$$

Example: For the symmetric potential i.e. $V(-x) = V(x)$, prove that parity operator $\hat{\pi}$ commute

with Hamiltonian $\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$.

Solution: $[\hat{H}, \hat{\pi}] \psi(x) = \hat{H} \hat{\pi} \psi(x) - \hat{\pi} \hat{H} \psi(x) = \hat{H} \psi(-x) - \pi \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + V(x) \psi(x) \right)$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(-x)}{\partial (-x)^2} + V(x) \psi(-x) - \pi \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + V(x) \psi(x) \right)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(-x)}{\partial (-x)^2} + V(x) \psi(-x) - \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(-x)}{\partial (-x)^2} + V(-x) \psi(-x) \right)$$

$$\text{Hence } V(-x) = V(x) \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(-x)}{\partial x^2} + V(x) \psi(-x) \right) - \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(-x)}{\partial x^2} + V(x) \psi(-x) \right) = 0$$

$$[\hat{H}, \hat{\pi}] \psi(x) = 0 \Rightarrow [\hat{H}, \hat{\pi}] = 0$$

$$\hat{L}_z = \hat{X}\hat{P}_y - \hat{Y}\hat{P}_x, \text{ find the value of } [\hat{X}, \hat{L}_z] \text{ and } [\hat{P}_y, \hat{L}_z].$$

$$[\hat{X}, \hat{L}_z] = [\hat{X}, \hat{X}\hat{P}_y - \hat{Y}\hat{P}_x] = [\hat{X}, \hat{X}]\hat{P}_y + \hat{X}[\hat{X}, \hat{P}_y] - [\hat{X}, \hat{Y}]\hat{P}_x - \hat{Y}[\hat{X}, \hat{P}_x]$$

$$= 0 + 0 - 0 - i\hbar\hat{Y} = -i\hbar\hat{Y}$$

$$[\hat{P}_y, \hat{L}_z] = [\hat{P}_y, \hat{X}\hat{P}_y] - [\hat{P}_y, \hat{Y}\hat{P}_x] = 0 - (-i\hbar)\hat{P}_x = +i\hbar\hat{P}_x$$

Example: Given the usual canonical commutation relations, the commutator $[A, B]$ of $A = i(xp_y + yp_x)$ and $B = (yp_z + zp_y)$. Find the value of $[A, B]$.

Solution: $[A, B] = [(ixp_y + iyp_x), (yp_z + zp_y)]$

$$[A, B] = i[xp_y, yp_z] + i[yp_x, yp_z] + i[xp_y, zp_y] + i[yp_x, zp_y]$$

$$[A, B] = i[xp_y, yp_z] + 0 + 0 + i[yp_x, zp_y] = i[xp_y, yp_z] + i[yp_x, zp_y]$$

$$[A, B] = ix[p_y, yp_z] + i[x, yp_z]p_y + iy[p_x, zp_y] + i[y, zp_y]p_x$$

$$[A, B] = ix[p_y, yp_z] + 0 + 0 + i[y, zp_y]p_x = ix[p_y, yp_z] + i[y, zp_y]p_x$$

$$[A, B] = ix \times (-i\hbar)p_z + izi\hbar \times p_x$$

$$[A, B] = \hbar(xp_z - p_xz)$$

Example: Let x and p denote, respectively, the coordinate and momentum operators satisfying the canonical commutation relation $[x, p] = i$ in natural units ($\hbar = 1$). Then the commutator find commutator $[x, p \sin p]$ is

Solution: $\therefore [x, p] = i$

$$[x, p \sin p] = [x, p] \sin p + p[x, \sin p] = i \sin p + p \left[x, p - \frac{p^3}{3} \dots \right]$$

$$= i \sin p + p \left[[x, p] - \left[x, \frac{p^3}{3} \right] \dots \right] = i \sin p + p \left[i - \frac{3ip^2}{3} \dots \right]$$

$$\Rightarrow [x, p \sin p] = i \sin p + p \left[i - \frac{3ip^2}{3} \dots \right] = i \sin p + p \left[i + \frac{ip^2}{2} \dots \right] = i(\sin p + p \cos p)$$

Chapter 1

Tools of Quantum Mechanics

6. Probability Density

If x is any random variable then any function $f(x)$ can be treated as probability density if

$$\int_{-\infty}^{+\infty} f(x) dx = 1 \text{ means Area under probability density curve } f(x) \text{ is } 1$$

Probability such that system has value of x between a to b

$$\text{i.e. } P(a < x < b) = \frac{\int_a^b f(x) dx}{\int_{-\infty}^{+\infty} f(x) dx}$$

$$\text{average value of } x \text{ if probability density function is } f(x) \text{ is given by } \langle x \rangle = \frac{\int_{-\infty}^{+\infty} xf(x) dx}{\int_{-\infty}^{+\infty} f(x) dx}$$

average value of x^2 if probability density function is $f(x)$ is given by $\langle x^2 \rangle = \frac{\int_{-\infty}^{\infty} x^2 f(x) dx}{\int_{-\infty}^{\infty} f(x) dx}$

standard deviation is measurement of dispersion given by Δx i.e. $\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$ the variance is $(\Delta x)^2 \geq 0$ variance so for any random variable process $\langle x^2 \rangle \geq \langle x \rangle^2$

Example: The probability density is defined as $\begin{cases} f(x) = x, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$

(a) Find the Probability such that $0 < x < \frac{1}{2}$.

(b) Find $\langle x \rangle$, $\langle x^2 \rangle$ and Δx .

Solution: (a) $p\left(0 \leq x \leq \frac{1}{2}\right) = \frac{\int_0^{1/2} f(x) dx}{\int_0^1 f(x) dx} = \frac{\int_0^{1/2} x dx}{\int_0^1 x dx} = \frac{1}{4}$

$$\langle x \rangle = \frac{\int_0^1 x f(x) dx}{\int_0^1 f(x) dx} = \frac{\int_0^1 x x dx}{\int_0^1 x dx} = \frac{\frac{1}{2}}{\frac{1}{2}} = \frac{2}{3} \quad \text{and} \quad \langle x^2 \rangle = \frac{\int_0^1 x^2 f(x) dx}{\int_0^1 f(x) dx} = \frac{\int_0^1 x^2 x dx}{\int_0^1 x dx} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{2}{4} = \frac{1}{2}$$

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \Delta x = \sqrt{\frac{1}{2} - \frac{4}{9}} = \sqrt{\frac{1}{18}}$$

Example: $f(x) = \begin{cases} x \exp\left(-\frac{x}{\lambda}\right), & \text{for } x \geq 0 \\ 0, & \text{otherwise} \end{cases}$, where $\lambda > 0$

(a) Find then the probability that system has value $x \geq \lambda$ assume $\lambda > 0$

(b) Find $\langle x \rangle$, $\langle x^2 \rangle$ and most probable value of x

(c) Find the most probable value of x .

Solution: (a) $p(x \geq \lambda) = \frac{\int_{\lambda}^{\infty} \exp\left(-\frac{x}{\lambda}\right) dx}{\int_0^{\infty} \exp\left(-\frac{x}{\lambda}\right) dx} = \frac{1}{e}$

$$(b) \langle x \rangle = \frac{\int_{-\infty}^{\infty} x f(x) dx}{\int_{-\infty}^{\infty} f(x) dx} = \frac{\int_0^{\infty} x x \exp\left(-\frac{x}{\lambda}\right) dx}{\int_0^{\infty} x \exp\left(-\frac{x}{\lambda}\right) dx} = \frac{\lambda^3 \sqrt{3}}{\lambda^2 \sqrt{2}} = 2\lambda$$

$$\langle x^2 \rangle = \frac{\int_{-\infty}^{\infty} x^2 f(x) dx}{\int_{-\infty}^{\infty} f(x) dx} = \frac{\int_0^{\infty} x^2 \cdot x \exp\left(-\frac{x}{\lambda}\right) dx}{\int_0^{\infty} x \exp\left(-\frac{x}{\lambda}\right) dx} = \frac{\lambda^4 |4|}{\lambda^2 |2|} = 6\lambda^2$$

(c) For most probable value $\frac{df}{dx} = 0 \Rightarrow \frac{dx \exp\left(-\frac{x}{\lambda}\right)}{dx} = 0$

$$\Rightarrow \exp\left(-\frac{x}{\lambda}\right) + x\left(-\frac{1}{\lambda}\right)\exp\left(-\frac{x}{\lambda}\right) = 0 \Rightarrow \exp\left(-\frac{x}{\lambda}\right)\left(1 - \frac{x}{\lambda}\right) = 0$$

Hence $\exp\left(-\frac{x}{\lambda}\right) \neq 0$ so $\left(1 - \frac{x}{\lambda}\right) = 0 \Rightarrow x = \lambda$ we can check $\left.\frac{d^2 f}{dx^2}\right|_{x=\lambda} < 0$ so $x = \lambda$ is $f(x)$ will

be maximum $f(\lambda) = \frac{\lambda}{e}$

Chapter 2

Postulates of Quantum Mechanics

In classical mechanics, if we know classical state of system i.e. generalized coordinate q_i and conjugate momentum of system p_i we can analyse any physical quantity of system. The time evolution of system can be derived by Hamiltonian equation of motion. For one dimensional system $\left(\frac{\partial H}{\partial q_i}\right) = -\dot{p}_i$ and $\left(\frac{\partial H}{\partial p_i}\right) = \dot{q}$

Similarly, in quantum mechanics the quantum mechanical counterparts to these ideas are specified by postulates, which enable us to understand:

- (a) How a quantum state is described mathematically at a given time t ,
- (b) How to calculate the various physical quantities from this quantum state,
- (c) Knowing the system's state at a time t , how to find the state at any later time t' ; that is, how to describe the time evolution of a system.

The answers to these questions are provided by the following set of five mathematical hypothesis which are known as postulates of quantum mechanics

1. Postulates of Quantum Mechanics

Postulates-1: State of Quantum Mechanical System

A quantum mechanical state of system defined by $|\psi(r, t)\rangle$, which belong to Hilbert space H associated to the system.

If $|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle$ are state of system then any linear superposition of state will also a state of system.

Mathematically $|\psi\rangle = c_1|\psi_1\rangle + c_2|\psi_2\rangle + \dots c_n|\psi_n\rangle$ is also state of a system.

If $|\psi\rangle$ is state of system then $|\psi|^2 = \psi^*\psi$ can be treated as probability density $f(x, y, z)$ if

$$\int_{-\infty}^{+\infty} |\psi|^2 d\tau = 1 \quad (\text{The integration is over all space})$$

In other words if ψ is normalized then $|\psi|^2$ is equivalent to probability density.

Postulates-2: Physical measurement of state

Any physical quantity $\mathcal{A}$ is identified as observable, there is an operator $\hat{A}$ associated to observable $\mathcal{A}$. If $\hat{A}$ is measured on state $|\psi\rangle$ the physical measurable quantities are the eigen values of operator i.e. $\hat{A}|\phi_n\rangle = a_n|\phi_n\rangle$ where $|\phi_n\rangle$ is eigen vector of operator $\hat{A}$ correspond to eigen value a_n .

Hence physical measurable quantities are real then eigen value associated to operator $\hat{A}$ is must real which ensure $\hat{A}$ must be Hermitian $\hat{A}^\dagger = \hat{A}$.

One should choose eigen state of operator $\hat{A}$ must be orthonormal in nature such that they can make complete basis $\sum_n |\phi_n\rangle\langle\phi_n| = I$.

Every eigen state can be treated as state of system but it is not necessary that every state is eigen state. If $\hat{A}$ is measurement on state $|\psi\rangle$ which means $|\psi\rangle$ can be written in basis of eigen vectors of $\hat{A}$ as $|\psi\rangle = I|\psi\rangle \Rightarrow |\psi\rangle = \sum_n |\phi_n\rangle\langle\phi_n|\psi\rangle$

Postulates-3: Probabilistic measurement of eigen value a_n on state $|\psi\rangle$

In quantum mechanics the measurement of any physical quantities are nor deterministic so there probability associated to each measurable quantity. From postulates 2 we know $\hat{A}|\phi_n\rangle = a_n|\phi_n\rangle$. If

operator $\hat{A}$ will measure on state $|\psi\rangle$ the probability associated with measurement is $P(a_n)$

mathematically $P(a_n) = \frac{|\langle \phi_n | \psi \rangle|^2}{\langle \psi | \psi \rangle}$

For degenerate eigen state $P(a_n) = \frac{\sum_j |\langle \phi_n^j | \psi \rangle|^2}{\langle \psi | \psi \rangle}$ where j is degeneracy of eigen value a_n and $|\phi_n^j\rangle$ are eigen state.

Basic Examples based on postulates 1, 2, 3

(a) If operator $\hat{A}$ is defined as $\hat{A} = \begin{bmatrix} a_0 & 0 \\ 0 & 2a_0 \end{bmatrix}$ is measured on $|\psi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ what is measurement and what will associated probability.

(b) If operator $\hat{A}$ is defined as $\hat{A} = \begin{bmatrix} a_0 & 0 \\ 0 & 2a_0 \end{bmatrix}$ is measured on $|\psi_2\rangle = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ what is measurement and what will associated probability.

Solution: According to postulates 2 the measurement is eigen value of operator

$$\hat{A}|\phi_n\rangle = a_n|\phi_n\rangle \text{ where } \hat{A} = \begin{bmatrix} a_0 & 0 \\ 0 & 2a_0 \end{bmatrix}, a_1 = a_0, |\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } a_2 = 2a_0, |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

(a) The state is $|\psi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ fortunately it is also eigen state corresponds to eigen value $a_1 = a_0$

from postulates 3 we can measure associated probability as

$$P(a_1) = \frac{|\langle \phi_1 | \psi_1 \rangle|^2}{\langle \psi_1 | \psi_1 \rangle} \Rightarrow P(a_0) = \frac{\left| \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2}{\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}} = 1$$

$$P(a_2) = \frac{|\langle \phi_2 | \psi_1 \rangle|^2}{\langle \psi_1 | \psi_1 \rangle} = P(2a_0) = \frac{\left| \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2}{\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}} = 0$$

Note this is very trivial result because eigen state is also state of system

(b) For $|\psi_2\rangle = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ from postulates 2 the measurement is eigen value

$$\hat{A}|\phi_n\rangle = a_n|\phi_n\rangle \text{ where } \hat{A} = \begin{bmatrix} a_0 & 0 \\ 0 & 2a_0 \end{bmatrix}, a_1 = a_0, |\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } a_2 = 2a_0, |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

from postulates 3 we can measure associated probability as

$$P(a_1) = \frac{|\langle\phi_1|\psi_2\rangle|^2}{\langle\psi_2|\psi_2\rangle} = P(a_0) = \frac{\left| \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right|^2}{\begin{pmatrix} 2 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix}} = \frac{4}{13}$$

$$P(a_2) = \frac{|\langle\phi_2|\psi_2\rangle|^2}{\langle\psi_2|\psi_2\rangle} = P(2a_0) = \frac{\left| \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right|^2}{2^2 + 3^2} = \frac{9}{13}$$

Another way we can write state $|\psi_2\rangle$ can be represented as orthonormal basis of eigen state $|\phi_1\rangle$ and $|\phi_2\rangle$ as $|\psi_2\rangle = 2|\phi_1\rangle + 3|\phi_2\rangle$ (we can use completeness relation)

$$P(a_1 = a_0) = \frac{|\langle\phi_1|\psi_2\rangle|^2}{\langle\psi_2|\psi_2\rangle} = \frac{2^2}{2^2 + 3^2} = \frac{4}{13} \text{ and } P(a_2 = 2a_0) = \frac{|\langle\phi_2|\psi_2\rangle|^2}{\langle\psi_2|\psi_2\rangle} = \frac{9}{13}$$

Which means if $\hat{A}$ will measure on state $|\psi_2\rangle$ one can get measurement a_0 or $2a_0$ with respective probability $\frac{4}{13}$ and $\frac{9}{13}$. But nobody can predict what will be next measurement on basis of present measurement.

Postulates-4: State just after measurement of eigen value a_n

If $\hat{A}$ is measured in state $|\psi\rangle$ eigen value a_n is the measurement at the instantaneous point of time State $|\psi\rangle$ will collapse (projected) in a direction of eigen state $|\phi_n\rangle$ corresponding to eigen value a_n .

$$|\psi\rangle \text{ after measurement of } a_n \text{ is } \frac{P|\psi\rangle}{\sqrt{\langle\psi|P|\psi\rangle}} = \frac{|\phi_n\rangle\langle\phi_n|\psi\rangle}{\sqrt{\langle\psi|\phi_n\rangle\langle\phi_n|\psi\rangle}} = \frac{|\phi_n\rangle\langle\phi_n|\psi\rangle}{\sqrt{(\langle\phi_n|\psi\rangle)^2}} = |\phi_n\rangle$$

If another operator $\hat{B}$ will measure just after measurement of A on state $|\psi\rangle$, $\hat{B}$ will measure eigen state $|\phi_n\rangle$. After time will evolve during the next measurement any of the eigen value of operator can be measured (according to postulate 2).

Example: The operator $\hat{A}$ is defined as $\hat{A}|\phi_n\rangle = na_0|\phi_n\rangle$ $n=1,2,3,\dots$ the state is given as $|\psi\rangle = 2|\phi_1\rangle + 3|\phi_2\rangle + 4|\phi_3\rangle$. Another operator $\hat{B}$ is defined as $\hat{B}|\phi_n\rangle = n^2b_0|\phi_n\rangle$ $n=1,2,3,\dots$

(a) After measurement of $\hat{A}$ when outcome is $2a_0$ at the same time $\hat{B}$ will be measured on state $|\psi\rangle$, what will be the measurement.

(b) After measurement of $\hat{A}$ when outcome is $2a_0$ after time evolved $\hat{B}$ will be measured on state $|\psi\rangle$, what will be the measurement.

Solution: (a) After measurement of $\hat{A}$ on $|\psi\rangle$ the measurement is $2a_0$ means at the time of when we get outcome $2a_0$, $|\psi\rangle$ is projected in direction of $|\phi_2\rangle$ at the same time if $\hat{B}$ will be measured actually on $|\phi_2\rangle$ which is also eigen state of $\hat{B}$ so measurement is $2^2b_0 = 4b_0$ only. This is outcome according to postulates 4

(b) after measurement of $\hat{A}$ on $|\psi\rangle$ the measurement is $2a_0$ means at the time of when we get outcome $2a_0$, $|\psi\rangle$ is projected in direction of $|\phi_2\rangle$ but $\hat{B}$ will be measured after time is evolved so according to postulates 2 again state will be in original state $|\psi\rangle = 2|\phi_1\rangle + 3|\phi_2\rangle + 4|\phi_3\rangle$ so can outcome $1^2b_0 = b_0$ or $2^2b_0 = 4b_0$ or $3^2b_0 = 9b_0$

Postulates-5: Time evolution of state

In classical mechanics the time evolution of state can be analyzed by Hamiltonian equation of motion which will ultimately give force equation $\vec{F} = \frac{md^2\vec{r}}{dt^2}$. In quantum mechanics it is not applicable so we find another way named as Schrodinger wave time dependent equation given as

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

A **stationary state** is a quantum mechanics with all observables independent of time. It is an eigenvector of the Hamiltonian. This corresponds to a state with a single definite energy (instead of a Quantum superposition of different energies). It is also called **energy eigenvector**, **energy eigenstate**, **energy eigenfunction**, or **energy eigenket**.

Time independent Schrodinger wave equation is given by $\hat{H}|u_n\rangle = E_n|u_n\rangle$

Where $\hat{H}$ is Hamiltonian of system and E_n is energy eigenvalue with corresponding energy eigen state $|u_n\rangle$ $n=1,2,3,\dots,m,\dots,n$

$$H|u_n\rangle = \frac{-\hbar^2}{2m} \nabla^2 |u_n\rangle + V|u_n\rangle = E_n |u_n\rangle \text{ where } \langle u_n | u_m \rangle = \delta_{n,m} \text{ and } \sum_n |u_n\rangle \langle u_n| = I$$

$$\text{Any state } |\psi(t=0)\rangle = \sum_n |u_n\rangle \langle u_n | \psi \rangle$$

Analogy to solve time dependent Schrödinger wave equation

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi(x,t)$$

$$\psi(x,t) = F(t)u(x) \text{ is solution of } i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi(x,t) \text{ then}$$

$$i\hbar u(x) \frac{dF}{dt} = F(t) \left(-\frac{\hbar^2}{2m} \frac{d^2 u}{dx^2} \right) + F(t)V(x)u(x) \text{ divided by } F(t)u(x) \text{ both side}$$

$$\frac{i\hbar}{F(t)} \frac{dF}{dt} = \frac{1}{u(x)} \left(\left(-\frac{\hbar^2}{2m} \frac{d^2 u}{dx^2} \right) + V(x)u(x) \right) \Rightarrow \frac{i\hbar}{F(t)} \frac{dF}{dt} = \frac{\hat{H}u(x)}{u(x)}$$

$$\frac{i\hbar}{F(t)} \frac{dF}{dt} = \frac{\hat{H}u(x)}{u(x)} \text{ give guarantee that } \frac{i\hbar}{F(t)} \frac{dF}{dt} = \frac{\hat{H}u(x)}{u(x)} = C$$

where C is constant which is identified as total energy E of system and $u(x)$ is identified as corresponding energy eigen state such that $\hat{H}u(x) = Eu(x)$.

$$\frac{i\hbar}{F(t)} \frac{dF}{dt} = E \Rightarrow \int \frac{dF}{F} = \frac{1}{i\hbar} \int dt \Rightarrow \ln F(t) = \frac{Et}{i\hbar} + \ln A \Rightarrow F(t) = A \exp\left(-\frac{iEt}{\hbar}\right)$$

So $\psi(x,t) = A \exp\left(\frac{-iEt}{\hbar}\right) u(x)$, where $u(x)$ is the energy eigen state with energy eigen value E

Similarly, we can verify in discrete basis

$$|\psi(t=0)\rangle = \sum_n |u_n\rangle \langle u_n | \psi \rangle$$

$$|\psi(t=t)\rangle = \sum_n \exp\left(\frac{-iE_n t}{\hbar}\right) |u_n\rangle \langle u_n | \psi \rangle$$

$$\text{We can verify } i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = \sum_n i(-iE_n) \exp\left(\frac{-iE_n t}{\hbar}\right) |u_n\rangle \langle u_n | \psi \rangle$$

$$\Rightarrow \sum_n E_n \exp\left(\frac{-iE_n t}{\hbar}\right) |u_n\rangle \langle u_n | \psi \rangle \Rightarrow \sum_n \exp\left(\frac{-iE_n t}{\hbar}\right) \hat{H} |u_n\rangle \langle u_n | \psi \rangle$$

$$\Rightarrow \hat{H} |\psi(t)\rangle = i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} \text{ which is popular Schrödinger wave equation, which is equivalent to}$$

Newton's law of motion in quantum mechanics

The solution of Schrödinger wave equation is $|\psi(t)\rangle = \left(\exp -\frac{i\hat{H}t}{\hbar} \right) |\psi(0)\rangle$

we know $\Rightarrow \langle \psi(t) | = \langle \psi(0) | \left(\exp \frac{i\hat{H}^\dagger t}{\hbar} \right)$

$\hat{H}^\dagger = \hat{H}$ so $\langle \psi(t) | = \langle \psi(0) | \left(\exp \frac{i\hat{H}t}{\hbar} \right)$

Evolution of norm with time

$$\langle \psi(t) | \psi(t) \rangle = \langle \psi(0) | \exp \left(\frac{i\hat{H}t}{\hbar} \right) \exp \left(-\frac{i\hat{H}t}{\hbar} \right) |\psi(0)\rangle = \langle \psi(0) | \exp \frac{(i\hat{H}t - i\hat{H}t)}{\hbar} \exp \frac{it[\hat{H}, -\hat{H}]}{2} |\psi(0)\rangle$$

$$\exp \frac{(i\hat{H}t - i\hat{H}t)}{\hbar} = \exp \hat{O} = I \text{ and } \exp \frac{it[\hat{H}, -\hat{H}]}{2} = \exp \hat{O} = I$$

$$\langle \psi(t) | \psi(t) \rangle = \langle \psi(0) | I I |\psi(0)\rangle = \langle \psi(0) | \psi(0) \rangle$$

We can conclude the state will evolve with time but its norms will preserve during the evolution, which means norm of the state will preserve and independent of time.

If we represent any state in orthonormal basis of energy eigen state as $|\psi(0)\rangle = \sum_n c_n |u_n\rangle$ where

$c_n = \langle u_n | \psi \rangle$ then it will fulfill two purposes.

(a) We can easily measure associated energy eigen value E_n with energy eigenstate $|u_n\rangle$ as we know $\hat{H}|u_n\rangle = E_n|u_n\rangle$

(b) We can analyze time evolution of state $|\psi(t)\rangle = \sum_n c_n \exp \left(\frac{-iE_n t}{\hbar} \right) |u_n\rangle$

The solution of Schrodinger equation must be

- (a) Single valued and value must be finite
- (b) Continuous
- (c) Differentiable
- (d) Square integrable.

Time Evolution of Probability Density

The Hamiltonian is defined as $H|\phi_n\rangle = E_n|\phi_n\rangle$ and $\hat{H}|\phi_m\rangle = E_m|\phi_m\rangle$

It is given as $\delta_{nm} = \langle \phi_n | \phi_m \rangle$

at $t = 0$ the wave function is $\psi(0) = c_n \phi_n + c_m \phi_m$ so $\psi^*(0) = c_n^* \phi_n^* + c_m^* \phi_m^*$

The probability density is given as $|\psi|^2 = |c_n|^2 |\phi_n|^2 + |c_m|^2 |\phi_m|^2 + 2 \operatorname{Re} c_n^* c_m \phi_n^* \phi_m$

After time t state will be $\psi(t) = c_n \exp\left(-\frac{iE_n t}{\hbar}\right) \phi_n + c_m \exp\left(-\frac{iE_m t}{\hbar}\right) \phi_m$

$$\psi^*(t) = c_n^* \exp\left(\frac{iE_n t}{\hbar}\right) \phi_n^* + c_m^* \exp\left(\frac{iE_m t}{\hbar}\right) \phi_m^*$$

$$\begin{aligned} |\psi(t)|^2 &= |c_n|^2 |\phi_n|^2 + |c_m|^2 |\phi_m|^2 + c_n^* \phi_n^* c_m \phi_m e^{\frac{it(E_n - E_m)}{\hbar}} + c_m^* \phi_m^* c_n \phi_n e^{-\frac{it(E_n - E_m)}{\hbar}} \\ &= |c_n|^2 |\phi_n|^2 + |c_m|^2 |\phi_m|^2 + 2 \operatorname{Re} c_n^* c_m \phi_n^* \phi_m \cdot \cos\left(\frac{t(E_n - E_m)}{\hbar}\right) \end{aligned}$$

The condition $|\psi(t)|^2 = |\psi(0)|^2 \Rightarrow \cos \frac{t(E_n - E_m)}{\hbar} = 1 \Rightarrow \frac{t(E_n - E_m)}{\hbar} = 2\pi$

system back to initial $t = \frac{2\pi\hbar}{|E_n - E_m|}$, which is also uncertainty in measure of time, $\Delta t = \frac{2\pi\hbar}{|E_n - E_m|}$

Average value of operator $\hat{A}$

$|\psi\rangle$ the operator $\hat{A}$ is defined as $\hat{A}|\phi_n\rangle = a_n |\phi_n\rangle$.

A state of system is given by $|\psi\rangle$ where $P(a_n)$ is probability of measurement on state $|\psi\rangle$ is given

$$\text{by } P(a_n) = \frac{|\langle \phi_n | \psi \rangle|}{\langle \psi | \psi \rangle}$$

The expectation value of $\hat{A}$ on State $|\psi\rangle$ which is mean value of $\hat{A}$

$$\text{i.e. } \langle \hat{A} \rangle = \frac{\langle \psi | \hat{A} | \psi \rangle}{\langle \psi | \psi \rangle} = \sum_n a_n P(a_n) = \frac{\int_{-\infty}^{+\infty} \psi^*(x) \hat{A}(x) \psi(x) dx}{\int_{-\infty}^{+\infty} \psi^* \psi dx}$$

The expectation value of $\hat{A}^2$ on State $|\psi\rangle$

$$\langle \hat{A}^2 \rangle = \frac{\langle \psi | \hat{A}^2 | \psi \rangle}{\langle \psi | \psi \rangle} = \sum_n a_n^2 P(a_n) = \frac{\int_{-\infty}^{+\infty} \psi^*(x) \hat{A}^2 \psi(x) dx}{\int_{-\infty}^{+\infty} \psi^* \psi dx}$$

The error in measurement of $\Delta \hat{A} = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2}$ $\langle \hat{A}^2 \rangle \geq \langle \hat{A} \rangle^2$

$\langle A \rangle$ is equivalent to classical result of measurement.

Example: If eigen value of operator $\hat{A}$ is 0, $2a_0$, $2a_0$ and corresponding normalized eigen vector

is $\frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}$, $\frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ i \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ respectively t system is in state $\frac{1}{6}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$ then

(a) When $\hat{A}$ is measured on system in state $\frac{1}{6}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$ then what is probability to getting value 0, $2a_0$, respectively.

(b) What is the expectation value of $\hat{A}$?

Solution: For eigen value $\lambda_1 = 0$ the eigen vector is $|\phi_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}$,

$$P(0) = \frac{|\langle \phi_1 | \psi(t) \rangle|^2}{\langle \psi | \psi \rangle} = \frac{8}{17}$$

$\lambda_2 = \lambda_3 = 2a_0$ i.e., $\lambda = 2a_0$ is doubly degenerate so two orthonormal eigen states are $|\phi_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ i \\ 1 \end{pmatrix}$

and $|\phi_3\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$$P(2a_0) = \frac{|\langle \phi_2 | \psi(t) \rangle|^2}{\langle \psi | \psi \rangle} + \frac{|\langle \phi_3 | \psi(t) \rangle|^2}{\langle \psi | \psi \rangle} = \frac{\frac{2}{9}}{\frac{17}{17}} + \frac{1}{17} = \frac{9}{17}, \quad \langle A \rangle = 0 \times \frac{8}{17} + 2a_0 \times \frac{9}{17} = \frac{18a_0}{17}$$

Example: Operator $\hat{A}$ is defined as $\hat{A} = \begin{pmatrix} 0 & a_0 \\ a_0 & 0 \end{pmatrix}$. If A will measure on $|\psi\rangle = \begin{pmatrix} 3 \\ 4i \end{pmatrix}$

(i) What will be the measurement with what probability?

(ii) If $|\phi_1\rangle$ and $|\phi_2\rangle$ are orthogonal eigen state of A . Prove that $|\phi_1\rangle$ and $|\phi_2\rangle$ will make a complete basis

(iii) Write down $|\psi\rangle$ in the basis of $|\phi_1\rangle$ and $|\phi_2\rangle$. Verify your result.

Solution: (i) For eigen value $|\hat{A} - \lambda I| = 0 \Rightarrow \begin{vmatrix} -\lambda & a_0 \\ a_0 & -\lambda \end{vmatrix} = 0 \Rightarrow (\lambda^2 - a_0^2) = 0 \Rightarrow \lambda_1 = a_0, \lambda_2 = -a_0$

So, the measurement is $\lambda_1 = a_0$ and $\lambda_2 = -a_0$.

Now let $|\phi_1\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ will be the eigen state corresponding to $a_1 = a_0$

$$\hat{A}|\phi_1\rangle = a_0|\phi_1\rangle \Rightarrow \begin{pmatrix} 0 & a_0 \\ a_0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = a_0 \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \Rightarrow a_1 = a_2 \Rightarrow |\phi_1\rangle = \begin{pmatrix} a_1 \\ a_1 \end{pmatrix}$$

We normalized $|\phi_1\rangle$ to get a single state $\langle\phi_1|\phi_1\rangle = 1$

$$\begin{pmatrix} a_1^* & a_1^* \end{pmatrix} \begin{pmatrix} a_1 \\ a_1 \end{pmatrix} = 1 \Rightarrow 2a_1^2 = 1 \Rightarrow a_1 = \frac{1}{\sqrt{2}} \Rightarrow |\phi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For the eigen value $a_2 = -a_0$, $|\phi_2\rangle = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$

$$\hat{A}|\phi_2\rangle = -a_0|\phi_2\rangle \Rightarrow \begin{pmatrix} 0 & a_0 \\ a_0 & 0 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = -a_0 \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \Rightarrow b_1 = -b_2 \Rightarrow |\phi_2\rangle = \begin{pmatrix} b_1 \\ -b_1 \end{pmatrix}$$

$$\langle\phi_2|\phi_2\rangle = 1 \Rightarrow \begin{pmatrix} b_1^* & -b_1^* \end{pmatrix} \begin{pmatrix} b_1 \\ -b_1 \end{pmatrix} = 1 \Rightarrow b_1 = \frac{1}{\sqrt{2}} \Rightarrow |\phi_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$P(a_0) = \frac{|\langle\phi_1|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \frac{\left| \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 3 \\ 4i \end{pmatrix} \right|^2}{(3-4i) \begin{pmatrix} 3 \\ 4i \end{pmatrix}} = \frac{\left| \frac{3}{\sqrt{2}} + i \frac{4}{\sqrt{2}} \right|^2}{3^2 + 4^2} = \frac{\left| \frac{3}{\sqrt{2}} \right|^2 + \left| \frac{4}{\sqrt{2}} \right|^2}{3^2 + 4^2} = \frac{\frac{25}{2}}{25} = \frac{1}{2}$$

$$P(-a_0) = \frac{|\langle\phi_2|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \frac{\left| \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 3 \\ 4i \end{pmatrix} \right|^2}{(3-4i) \begin{pmatrix} 3 \\ 4i \end{pmatrix}} = \frac{\left| \frac{3}{\sqrt{2}} - i \frac{4}{\sqrt{2}} \right|^2}{3^2 + 4^2} = \frac{\left| \frac{3}{\sqrt{2}} \right|^2 + \left| \frac{-4}{\sqrt{2}} \right|^2}{25} = \frac{\frac{25}{2}}{25} = \frac{1}{2}$$

$$P(a_0) + P(-a_0) = 1$$

$$(ii) |\phi_1\rangle\langle\phi_1| + |\phi_2\rangle\langle\phi_2| = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} + \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1 \text{ so } |\phi_1\rangle \text{ and } |\phi_2\rangle \text{ will make a complete basis}$$

$$(iii) |\psi\rangle = \{\langle\phi_1|\psi\rangle\}|\phi_1\rangle + \{\langle\phi_2|\psi\rangle\}|\phi_2\rangle = \frac{3+4i}{\sqrt{2}}|\phi_1\rangle + \frac{3-4i}{\sqrt{2}}|\phi_2\rangle$$

$$P(a_0) = \frac{|\langle\phi_1|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \frac{\left| \frac{3+4i}{\sqrt{2}} \right|^2}{\frac{1}{2}(3^2+4^2) + \frac{1}{2}(3^2+4^2)} = \frac{\left(\frac{3}{\sqrt{2}} \right)^2 + \left(\frac{4}{\sqrt{2}} \right)^2}{3^2+4^2} = \frac{1}{2}$$

$$P(-a_0) = \frac{|\langle\phi_2|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \frac{\frac{1}{2}|3-4i|^2}{3^2+4^2} = \frac{1}{2} \left(\frac{3^2+4^2}{3^2+4^2} \right) = \frac{1}{2} \Rightarrow P(a_0) + P(-a_0) = 1$$

Example: An operator $\hat{A}$ is given by $\hat{A} = \begin{pmatrix} a_0 & 0 & 0 \\ 0 & a_0 & 0 \\ 0 & 0 & -a_0 \end{pmatrix}$ If $\hat{A}$ will measured on a state

$|\psi\rangle = \begin{pmatrix} 3 \\ 4i \\ 5 \end{pmatrix}$, then what will be the measurement with what probability?

Solution: As here A is a diagonal matrix the eigen value of A will be

$$\lambda_1 = a_0, \lambda_2 = a_0, \lambda_3 = -a_0 \text{ Corresponding eigen state is } |\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |\phi_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Now the probability of degenerate eigen value

$$P(a_0) = \frac{|\langle \phi_1 | \psi \rangle|^2}{\langle \psi | \psi \rangle} \oplus \frac{|\langle \phi_2 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{|3|^2}{|3|^2 + |4i|^2 + |5|^2} \oplus \frac{|4i|^2}{|3|^2 + |4i|^2 + |5|^2} = \frac{9}{50} + \frac{16}{50} = \frac{25}{50} = \frac{1}{2}$$

$$P(-a_0) = \frac{|\langle \phi_3 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{\left| \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 4i \\ 5 \end{pmatrix} \right|^2}{\begin{pmatrix} 3 & -4i & 5 \end{pmatrix} \begin{pmatrix} 3 \\ 4i \\ 5 \end{pmatrix}} = \frac{5^2}{3^2 + 4^2 + 25} = \frac{25}{50} = \frac{1}{2}$$

$$\therefore P(-a_0) + P(a_0) = 1$$

Example: $\hat{A} = \begin{bmatrix} a_0 & 0 & 0 \\ 0 & 0 & a_0 \\ 0 & a_0 & 0 \end{bmatrix}$. If A will measured on a state $|\psi\rangle = \begin{pmatrix} 3 \\ 4i \\ 5 \end{pmatrix}$, what will be

measurement and what is probability?

Solution: From eigen value

$$|\hat{A} - \lambda I| = 0 \Rightarrow \begin{vmatrix} a_0 - \lambda & 0 & 0 \\ 0 & -\lambda & a_0 \\ 0 & a_0 & -\lambda \end{vmatrix} = 0 \Rightarrow (a_0 - \lambda)(\lambda^2 - a_0^2) = 0$$

$$(a_0 - \lambda)(\lambda + a_0)(\lambda - a_0) = 0 \Rightarrow \lambda_1 = -a_0, \lambda_2 = a_0, \lambda_3 = a_0,$$

The measurement is $-a_0$ or a_0

For $\lambda_1 = -a_0$ Let the eigen state be $|\phi_1\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$

$$A|\phi_1\rangle = -a_0|\phi_1\rangle \Rightarrow \begin{pmatrix} a_0 & 0 & 0 \\ 0 & 0 & a_0 \\ 0 & a_0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = -a_0 \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

$$a_1 = -a_1 \Rightarrow a_1 = 0$$

$$a_3 \cdot a_0 = -a_2 a_0 \Rightarrow a_2 = -a_3 \text{ and } a_2 a_0 = -a_3 a_0 \Rightarrow a_2 = -a_3 \text{ so } |\phi_1\rangle = \begin{pmatrix} 0 \\ a_2 \\ -a_2 \end{pmatrix}$$

using normalization condition $\langle \phi_1 | \phi_1 \rangle = 1$

$$\begin{pmatrix} 0 & a_2^* & -a_2^* \end{pmatrix} \begin{pmatrix} 0 \\ a_2 \\ -a_2 \end{pmatrix} = 1 \Rightarrow a_2^2 + a_2^2 = 1 \Rightarrow a_2 = \frac{1}{\sqrt{2}} \Rightarrow |\phi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

Now we will calculate eigen vector corresponds to for $\lambda_2 = a_0$. Let corresponding eigen state be

$$|\phi_2\rangle = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$\hat{A}|\phi_2\rangle = a_0|\phi_2\rangle \Rightarrow \begin{pmatrix} a_0 & 0 & 0 \\ 0 & 0 & a_0 \\ 0 & a_0 & 0 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = a_0 \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$b_1 a_0 = b_1 a_0 \Rightarrow b_1 = b_1, b_3 a_0 = b_2 a_0 \Rightarrow b_2 = b_3, b_2 a_0 = b_3 a_0 \Rightarrow b_2 = b_3, \therefore |\phi_2\rangle = \begin{pmatrix} b_1 \\ b_2 \\ b_2 \end{pmatrix}$$

We choose any arbitrary value $b_1 = 0$ to make $|\phi_2\rangle$

$$|\phi_2\rangle = \begin{pmatrix} 0 \\ b_2 \\ b_2 \end{pmatrix}. \text{ Using normalisation condition } \langle \phi_2 | \phi_2 \rangle = 1 \Rightarrow \begin{pmatrix} 0 & b_2^* & b_2^* \end{pmatrix} \begin{pmatrix} 0 \\ b_2 \\ b_2 \end{pmatrix} = 1$$

$$2b_2^2 = 1 \Rightarrow b_2 = \frac{1}{\sqrt{2}} \Rightarrow |\phi_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\text{For } \lambda_3 = a_0 \hat{A}|\phi_3\rangle = a_0|\phi_3\rangle \Rightarrow \begin{pmatrix} a_0 & 0 & 0 \\ 0 & 0 & a_0 \\ 0 & a_0 & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = a_0 \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$\text{Now we find } \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \text{ in the form of } |\phi_3\rangle = \begin{pmatrix} c_1 \\ c_2 \\ c_2 \end{pmatrix} \text{ such that } \langle \phi_2 | \phi_3 \rangle = 0$$

$$\begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_2 \end{pmatrix} = 0 \Rightarrow c_2 = 0$$

from normalization condition $\langle \phi_3 | \phi_3 \rangle = 1 \Rightarrow c_1 = 1$ so $|\phi_3\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$$P(-a_0) = \frac{|\langle \phi_1 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{\left| \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 3 \\ 4i \\ 5 \end{pmatrix} \right|^2}{(3 - 4i) \begin{pmatrix} 3 \\ 4i \\ 5 \end{pmatrix}} = \frac{\frac{1}{2} |4i - 5|^2}{|3|^2 + |4i|^2 + |3|^2} = \frac{1}{2} \frac{|4|^2 + |5|^2}{50} = \frac{41}{100}$$

$$P(a_0) = \frac{|\langle \phi_2 | \psi \rangle|^2}{\langle \psi | \psi \rangle} \oplus \frac{|\langle \phi_3 | \psi \rangle|^2}{\langle \psi | \psi \rangle}$$

$$\frac{\left| \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 3 \\ 4i \\ 5 \end{pmatrix} \right|^2}{|3|^2 + |4i|^2 + |5|^2} \oplus \frac{\left| \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 4i \\ 5 \end{pmatrix} \right|^2}{|3|^2 + |4i|^2 + |5|^2} = \frac{3}{2} \frac{|4i|^2 + |5|^2}{50} \oplus \frac{|3|^2}{50} = \frac{41}{100} + \frac{9}{50} = \frac{59}{100}$$

$$\therefore P(-a_0) + P(a_0) = \frac{41}{100} + \frac{41}{100} + \frac{9}{50} = 1$$

Example: Operation $\hat{A}$ is defined as $\hat{A}|\phi_n\rangle = n^2 a_0 |\phi_n\rangle$ where $n = 1, 2, 3, \dots$ it is also given

$$\langle \phi_m | \phi_n \rangle = \delta_{n,m}$$

Any state $|\psi\rangle$ at $t = 0$ is given as $|\psi\rangle = 3|\phi_1\rangle - 2i|\phi_2\rangle + (3 + 4i)|\phi_4\rangle$

If $\hat{A}$ will measured on the state $|\psi\rangle$ then what will be the measurement with what probability?

Solution: When $\hat{A}$ will measured on $|\phi_n\rangle$ it will give the eigen value $n^2 a_0$

$$\hat{A}|\phi_n\rangle = n^2 a_0 |\phi_n\rangle, A|\phi_1\rangle = 1^2 a_0 |\phi_1\rangle, A|\phi_2\rangle = 2^2 a_0 |\phi_2\rangle, A|\phi_4\rangle = 4^2 a_0 |\phi_4\rangle$$

So, that measurement is $a_0, 4a_0, 16a_0$ $|\psi\rangle = 3|\phi_1\rangle - 2i|\phi_2\rangle + (3 + 4i)|\phi_4\rangle$

Here $|\psi\rangle$ is given in terms of $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle$

$$P(a_0) = \frac{|\langle \phi_1 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{|3|^2}{|3|^2 + |-2i|^2 + |3 + 4i|^2} = \frac{|3|^2}{9 + 4 + 9 + 16} = \frac{9}{38}$$

$$P(4a_0) = \frac{|\langle \phi_2 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{|-2i|^2}{38} = \frac{4}{38}, P(16a_0) = \frac{|\langle \phi_4 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{|3 + 4i|^2}{38} = \frac{3^2 + 4^2}{38} = \frac{25}{38}$$

$$P(a_0) + P(4a_0) + P(16a_0) = \frac{9}{38} + \frac{4}{38} + \frac{25}{38} = 1$$

Example: The Hamiltonian operator is defined as $\hat{H}|\phi_n\rangle = E_n|\phi_n\rangle$, where $n = 1, 2, 3, \dots, m, \dots, n$ at $t = 0, |\psi\rangle = c_n|\phi_n\rangle + c_m|\phi_m\rangle$ $\langle\phi_n|\phi_m\rangle = \delta_{nm}$

(a) At $t = 0$, if H is measured on state $|\psi\rangle$, what will be measurement with what probability?

(b) Find the angular value of H and H^2 on state $|\psi\rangle$ at $t = 0$.

(c) After what time $t, |\psi(t)\rangle$ is orthogonal to $|\psi(0)\rangle$?

(d) After what time $t, |\psi(t)|^2 = |\psi(0)|^2$?

(e) Find the value of Δt if H is measured on $|\psi\rangle$ at $t = t_0$ what will be measurement with what probability?

(f) Find the angular value of H and H^2 on state $|\psi(t)\rangle$ where $t = t_0$.

Solution: (a) Measurement: at $t = 0$ $|\psi\rangle = c_n|\phi_n\rangle + c_m|\phi_m\rangle$

$\hat{H}|\phi_n\rangle = E_n|\phi_n\rangle$ and $\hat{H}|\phi_m\rangle = E_m|\phi_m\rangle$, so measurement is either E_n or E_m

$$P(E_n) = \frac{|c_n|^2}{|c_n|^2 + |c_m|^2}, \quad P(E_m) = \frac{|c_m|^2}{|c_n|^2 + |c_m|^2}$$

$$(b) \langle H \rangle = E_n P(E_n) + E_m P(E_m) = \frac{E_n |c_n|^2}{|c_n|^2 + |c_m|^2} + \frac{E_m |c_m|^2}{|c_n|^2 + |c_m|^2}$$

$$\langle \hat{H}^2 \rangle = E_n^2 P(E_n) + E_m^2 P(E_m) = \frac{E_n^2 |c_n|^2}{|c_n|^2 + |c_m|^2} + \frac{E_m^2 |c_m|^2}{|c_n|^2 + |c_m|^2}$$

we can solve it with using formula $\langle H \rangle = \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle}$

if $|\psi\rangle = c_n|\phi_n\rangle + c_m|\phi_m\rangle$ then $\langle \psi | = c_n^* \langle \phi_n | + c_m^* \langle \phi_m |$

$$H|\psi\rangle = c_n E_n |\phi_n\rangle + c_m E_m |\phi_m\rangle \Rightarrow \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{|c_n|^2 E_n + E_m |c_m|^2}{|c_n|^2 + |c_m|^2}$$

(c) $|\psi(0)\rangle = c_n|\phi_n\rangle + c_m|\phi_m\rangle$

$$|\psi(t)\rangle = c_n \exp\left(-\frac{iE_n t}{\hbar}\right) |\phi_n\rangle + c_m \exp\left(-\frac{iE_m t}{\hbar}\right) |\phi_m\rangle$$

$$\langle \psi(0) | \psi(t) \rangle = 0 \Rightarrow |c_n|^2 \exp\left(-\frac{iE_n t}{\hbar}\right) + |c_m|^2 \exp\left(-\frac{iE_m t}{\hbar}\right) = 0$$

$$= \frac{|c_n|^2}{|c_m|^2} = -\exp\left(\frac{i(E_n - E_m)t}{\hbar}\right) \Rightarrow \cos \frac{(E_n - E_m)t}{\hbar} = -\frac{|c_n|^2}{|c_m|^2}$$

$$t = \frac{\hbar}{|E_n - E_m|} \cos^{-1} \left(-\frac{|c_n|^2}{|c_m|^2} \right) \quad \text{where } 0 \leq \frac{|c_n|^2}{|c_m|^2} \leq 1$$

(d) $|\psi(t)|^2 = |\psi(0)|^2$

$$\cos \frac{(E_n - E_m)t}{\hbar} = 1 \Rightarrow t = \frac{2\pi\hbar}{|E_n - E_m|} \Rightarrow \Delta t \simeq \frac{2\pi\hbar}{|E_n - E_m|}$$

$$\begin{aligned} \text{(e)} \quad |\psi(t=t_0)\rangle &= c_n \exp\left(-\frac{iE_n t_0}{\hbar}\right) |\phi_n(0)\rangle + c_m \exp\left(-\frac{iE_m t_0}{\hbar}\right) |\phi_m\rangle \\ &= c'_n |\phi_n(0)\rangle + c'_m |\phi_m(0)\rangle \end{aligned}$$

$$P(E_n) = \frac{|c'_n|^2}{|c'_n|^2 + |c'_m|^2} = \frac{|c_n|^2}{|c_n|^2 + |c_m|^2} \quad \text{and} \quad P(E_m) = \frac{|c'_m|^2}{|c'_n|^2 + |c'_m|^2} = \frac{|c_m|^2}{|c_n|^2 + |c_m|^2}$$

(f) Hence after time t measurement of energy and associated probability is same as $t = 0$

$$\langle H \rangle = E_n P(E_n) + E_m P(E_m) = \frac{E_n |c_n|^2}{|c_n|^2 + |c_m|^2} + \frac{E_m |c_m|^2}{|c_n|^2 + |c_m|^2}$$

$$\langle \hat{H}^2 \rangle = E_n^2 P(E_n) + E_m^2 P(E_m) = \frac{E_n^2 |c_n|^2}{|c_n|^2 + |c_m|^2} + \frac{E_m^2 |c_m|^2}{|c_n|^2 + |c_m|^2}$$

Example: A state function is given by

$$|\psi\rangle = |\phi_1\rangle + \frac{1}{\sqrt{2}} |\phi_2\rangle$$

It is given that $\langle \phi_i | \phi_j \rangle = \delta_{ij}$

(a) check $|\psi\rangle$ is normalized or not

(b) write down normalized wavefunction $\langle \psi |$.

(c) It is given $\hat{H} |\phi_n\rangle = (n+1)\hbar\omega |\phi_n\rangle \quad n = 0, 1, 2, 3, \dots$

If $\hat{H}$ will measured on $|\psi\rangle$, what will be measurement with what probability.

(d) Find the expectation value at $\hat{H}$ i.e., $\langle \hat{H} \rangle$

(e) Find the error in the measurement in $\hat{H}$ i.e. $\Delta \hat{H}$.

Solution: (a) $|\psi\rangle = |\phi_1\rangle + \frac{1}{\sqrt{2}} |\phi_2\rangle$

To check normalization, one should verify. $\langle \psi | \psi \rangle = 1$

$$|\psi\rangle = |\phi_1\rangle + \frac{1}{\sqrt{2}} |\phi_2\rangle$$

$$\langle \psi | \psi \rangle = \langle \phi_1 | \phi_1 \rangle + \langle \phi_1 | \phi_2 \rangle \frac{1}{\sqrt{2}} \langle \phi_2 | \phi_1 \rangle + \frac{1}{\sqrt{2}} \langle \phi_2 | \phi_2 \rangle \left(\frac{1}{\sqrt{2}} \right) = 1 + 0 + 0 + \frac{1}{2} = \frac{3}{2}$$

The value of $\langle \psi | \psi \rangle = \frac{3}{2}$ so $|\psi\rangle$ is not normalized.

(b) Now we need to find normalized $|\psi\rangle$ let A be normalization constant.

$$|\psi\rangle = A \left(|\phi_1\rangle + \frac{1}{\sqrt{2}} |\phi_2\rangle \right)$$

$$\langle \psi | \psi \rangle = A^2 + \frac{A^2}{2} = 1 \Rightarrow \frac{3A^2}{2} = 1 \Rightarrow A = \sqrt{\frac{2}{3}}$$

So $|\psi\rangle = \sqrt{\frac{2}{3}} |\phi_1\rangle + \frac{1}{\sqrt{3}} |\phi_2\rangle$

$$\langle\psi| = \langle\phi_1| \sqrt{\frac{2}{3}} + \langle\phi_2| \frac{1}{\sqrt{3}}$$

(c) It is given that

$$\hat{H} |\phi_n\rangle = (n+1)\hbar\omega \quad n = 0, 1, 2, \dots$$

$$\hat{H} |\phi_1\rangle = 2\hbar\omega \quad \hat{H} |\phi_2\rangle = 3\hbar\omega$$

When $\hat{H}$ will be measured $|\psi\rangle$ it will be measured either $2\hbar\omega$ or $3\hbar\omega$

The probability of measuring $2\hbar\omega$ is $P(2\hbar\omega)$ is given by

$$P(2\hbar\omega) = \frac{|\langle\phi_1|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \frac{2}{3} \quad P(3\hbar\omega) = \frac{|\langle\phi_2|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \frac{1}{3}$$

So when $\hat{H}$ will measure state $|\psi\rangle$ the following outcome will come.

Measurement of $\hat{H}$ on state : $|\phi_1\rangle \quad |\phi_2\rangle$

Measurement : $2\hbar\omega \quad 3\hbar\omega$

Probability : $\frac{2}{3} \quad \frac{1}{3}$

(d) $\langle\hat{H}\rangle = \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle} = \sum_n P_n(E_n)E_n = 2\hbar\omega \times \frac{2}{3} + 3\hbar\omega \times \frac{1}{3} \Rightarrow \langle\hat{H}\rangle = \frac{7\hbar\omega}{3}$

$$\begin{aligned} \langle\hat{H}^2\rangle &= \frac{\langle\psi|\hat{H}^2|\psi\rangle}{\langle\psi|\psi\rangle} = \sum P_n(E_n)E_n^2 \Rightarrow \langle\hat{H}^2\rangle = \frac{2}{3} \times (2\hbar\omega)^2 + \frac{1}{3} \times (3\hbar\omega)^2 \\ &= \frac{8\hbar^2\omega^2}{3} + \frac{9\hbar^2\omega^2}{3} = \frac{17\hbar^2\omega^2}{3} \end{aligned}$$

(e) The error in measurement in $\hat{H}$ is given as

$$\Delta\hat{H} = \sqrt{\langle\hat{H}^2\rangle - \langle\hat{H}\rangle^2}, \quad \langle\hat{H}^2\rangle = \frac{17\hbar^2\omega^2}{3}, \quad \langle\hat{H}\rangle^2 = \left(\frac{7\hbar\omega}{3}\right)^2 = \frac{49\hbar^2\omega^2}{9}$$

$$\Delta\hat{H} = \sqrt{\frac{17}{3} - \frac{49}{9}}\hbar\omega \Rightarrow \Delta\hat{H} = \sqrt{\frac{51-49}{9}}\hbar\omega = \frac{\sqrt{2}}{3}\hbar\omega$$

Example: Operator $\hat{A}$ is defined as $\hat{A}|a_1\rangle = a_1|a_1\rangle$ and $\hat{A}|a_2\rangle = a_2|a_2\rangle$

If A will measured on a state $|\psi\rangle$ which is defined as $|\psi\rangle = 3|a_1\rangle + 4|a_2\rangle$ where $|a_1\rangle$ and $|a_2\rangle$ are orthonormal vectors

(i) What will be the measurement and what is probability?

(ii) If A will measured on a state $|\psi\rangle$ the measurement is a_2 . Just after the measurement of $\hat{A}$, again operator $\hat{B}$ will measured on a state, $|\psi\rangle$, where $\hat{B}$ is defined as $\hat{B}|b_1\rangle = b_1|b_1\rangle$, $\hat{B}|b_2\rangle = b_2|b_2\rangle$ where $|b_1\rangle$ and $|b_2\rangle$ are orthonormal vectors

It is also given that $|a_1\rangle = \frac{1}{\sqrt{2}}(|b_1\rangle + |b_2\rangle)$ and $|a_2\rangle = \frac{1}{\sqrt{2}}(|b_1\rangle - |b_2\rangle)$

Then what will be the measurement and what is probability.

(iii) After evolving time again $\hat{B}$ will measured on $|\psi\rangle$. What will measurement and what is probability?

Solution: (i) According to postulates 2 and postulates 2

$\hat{A}|a_1\rangle = a_1|a_1\rangle$ $\hat{A}|a_2\rangle = a_2|a_2\rangle$ The measurement is eigen value a_1 or a_2

$$P(a_1) = \frac{|\langle a_1 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{|3|^2}{|3|^2 + |4|^2} = \frac{9}{25} \text{ and } P(a_2) = \frac{|\langle a_2 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{|4|^2}{|3|^2 + |4|^2} = \frac{16}{25}$$

$$\therefore P(a_1) + P(a_2) = \frac{9}{25} + \frac{16}{25} = 1$$

(ii) According to postulates 4 if $\hat{A}$ will measured on a state $|\psi\rangle$ and the measurement is a_2 , that means $|\psi\rangle$ will projected in the direction of $|a_2\rangle$

Just after measurement of $\hat{A}$, if $\hat{B}$ will measure on the state $|\psi\rangle$ $\hat{B}$ will measured on eigen state corresponds to eigen value a_2 i.e. $|a_2\rangle$

As $\hat{B}|b_1\rangle = b_1|b_1\rangle$, $\hat{B}|b_2\rangle = b_2|b_2\rangle$ and $|a_2\rangle = \frac{1}{\sqrt{2}}(|b_1\rangle - |b_2\rangle)$

Now the measurement is b_1 or b_2

$$P(b_1) = \frac{|\langle b_1 | a_2 \rangle|^2}{\langle a_2 | a_2 \rangle} = \frac{\left| \frac{1}{\sqrt{2}} \right|^2}{\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2} = \frac{1}{2} \text{ and } P(b_2) = \frac{|\langle b_2 | a_2 \rangle|^2}{\langle a_2 | a_2 \rangle} = \frac{\left| \frac{1}{\sqrt{2}} \right|^2}{\frac{1}{2} + \frac{1}{2}} = \frac{1}{2}$$

(iii) When time is evolved sufficiently then the state become again $|\psi\rangle$. Now $\hat{B}$ will measured on $|\psi\rangle$

Now we write $|\psi\rangle$ in terms of $|b_1\rangle$ and $|b_2\rangle$

$$|\psi\rangle = 3|a_1\rangle + 4|a_2\rangle = \frac{3}{\sqrt{2}}(|b_1\rangle + |b_2\rangle) + \frac{4}{\sqrt{2}}(|b_1\rangle - |b_2\rangle)$$

$$|\psi\rangle = \frac{7}{\sqrt{2}}|b_1\rangle - \frac{1}{\sqrt{2}}|b_2\rangle$$

Here the measurement is again b_1 or b_2 as this is the eigen value.

$$P(b_1) = \frac{|\langle b_1 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{|7/\sqrt{2}|^2}{|7/\sqrt{2}|^2 + |1/\sqrt{2}|^2} = \frac{49}{50}$$

$$P(b_2) = \frac{|\langle b_2 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{|1/\sqrt{2}|^2}{|7/\sqrt{2}|^2 + |1/\sqrt{2}|^2} = \frac{1}{50}$$

$$P(b_1) + P(b_2) = 1$$

Example: A time $t = 0$ the state vector $|\psi(0)\rangle$ where $|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle + |\phi_2\rangle)$

It is given as Hamiltonian is defined as $\hat{H}|\phi_n\rangle = n^2 \epsilon_0 |\phi_n\rangle$ where $n = 1, 2, 3, \dots$ it is given as $\langle \phi_m | \phi_n \rangle = \delta_{m,n}$

(a) What is wave function $|\psi(t)\rangle$ at later time t .

(b) Write down expression of evolution of $|\psi(x, t)|^2$

(c) Find $\Delta\hat{H}$

(d) Find the value of $\Delta\hat{H} \cdot \Delta t$

Solution: (a) $|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left[\exp\left(\frac{-i\epsilon_0 t}{\hbar}\right) |\phi_1\rangle + \exp\left(\frac{-i4\epsilon_0 t}{\hbar}\right) |\phi_2\rangle \right]$

$$|\psi(t)\rangle \propto [|\phi_1\rangle + e^{-i\omega_{21}t} |\phi_2\rangle]$$

Where $\omega_{21} = \frac{E_2 - E_1}{\hbar} = \frac{3\epsilon_0}{\hbar}$

(b) Evolution of shape of the wave packet

$$|\psi(x, t)|^2 = \frac{1}{2} |\phi_1(x)|^2 + \frac{1}{2} |\phi_2(x)|^2 + \phi_1 \phi_2 \cos \omega_{21} t$$

(c) $\Delta\hat{H} = (\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2)^{1/2}$

$$\langle \hat{H} \rangle = \frac{1}{2} E_1 + \frac{1}{2} E_2 = \frac{1}{2} \epsilon_0 + \frac{1}{2} 4 \epsilon_0 = \frac{5}{2} \epsilon_0$$

$$\langle \hat{H}^2 \rangle = \frac{1}{2} E_1^2 + \frac{1}{2} E_2^2 = \frac{1}{2} (\epsilon_0)^2 + \frac{1}{2} (4 \epsilon_0)^2 = \frac{17}{2} \epsilon_0^2$$

$$\Delta \hat{H} = \left(\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2 \right)^{1/2} = \sqrt{\left(\frac{17}{2} - \frac{25}{4} \right)} \epsilon_0 = \sqrt{\frac{9}{4}} \epsilon_0 \Rightarrow \Delta \hat{H} = \frac{3}{2} \epsilon_0$$

$$(d) \quad \Delta \hat{H} = \frac{3}{2} \epsilon_0 \text{ and } \Delta t = \frac{2\pi\hbar}{E_2 - E_1} = \frac{2\pi\hbar}{3\epsilon_0} \text{ so } \Delta \hat{H} \cdot \Delta t = \frac{3}{2} \epsilon_0 \times \frac{2\pi\hbar}{3\epsilon_0} = \frac{h}{2}$$

Example: A state $|\psi\rangle$ is written in the form $|\psi(0)\rangle = \frac{1}{\sqrt{2}}|\psi_+\rangle + \frac{1}{\sqrt{2}}|\psi_-\rangle$

The energy eigen value is (E) and $(-E)$ associated to $|\psi_+\rangle$ and $|\psi_-\rangle$

(i) If H will measured on state $|\psi(0)\rangle$ then what will be the measurement and what probability?

(ii) After what time t , $|\psi(t)\rangle$ is orthogonal to $|\psi(0)\rangle$?

(iii) After what time $|\psi(t)|^2 = |\psi(0)|^2$

(iv) Write down the expression of state after time $t = \frac{h}{6E}$

(v) Find $\langle H \rangle$ and $\langle H^2 \rangle$ at time $t = \frac{h}{6E}$ on $|\psi(H)\rangle$.

(vi) Find the probability that the system will be in the state at $t = \frac{h}{6E}$ i.e., the probability that $|\psi(0)\rangle$ will be in the state $|\psi(H)\rangle$.

Solution: (i) A state $|\psi\rangle$ is written in the form $|\psi(0)\rangle = \frac{1}{\sqrt{2}}|\psi_+\rangle + \frac{1}{\sqrt{2}}|\psi_-\rangle$

If H will measured on state $|\psi(0)\rangle$ then the measurement is E or $-E$

$$P(E) = \frac{|\langle \psi_+ | \psi(0) \rangle|^2}{|\langle \psi_+ | \psi(0) \rangle|^2 + |\langle \psi_- | \psi(0) \rangle|^2} = \frac{\left| \frac{1}{\sqrt{2}} \right|^2}{\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2} = \frac{1}{2}$$

$$P(-E) = \frac{|\langle \psi_- | \psi(0) \rangle|^2}{|\langle \psi_+ | \psi(0) \rangle|^2 + |\langle \psi_- | \psi(0) \rangle|^2} = \frac{\left| \frac{1}{\sqrt{2}} \right|^2}{\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2} = \frac{1}{2}$$

$$(ii) \quad |\psi(0)\rangle = \frac{1}{\sqrt{2}}|\psi_+\rangle + \frac{1}{\sqrt{2}}|\psi_-\rangle \quad \langle \psi(0) | = \frac{1}{\sqrt{2}}\langle \psi_+ | + \frac{1}{\sqrt{2}}\langle \psi_- |$$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}|\psi_+\rangle \exp\left(\frac{-iEt}{\hbar}\right) + \frac{1}{\sqrt{2}}|\psi_-\rangle \exp\left(\frac{iEt}{\hbar}\right)$$

$$\langle\psi(0)|\psi(t)\rangle = 0 \Rightarrow t = \frac{\hbar}{|E_n - E_m|} \cos^{-1} \left\{ -\frac{|C_n|^2}{|C_m|^2} \right\} \text{ here } E_n = E, E_m = -E \text{ and } |C_n|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \text{ and}$$

$$|C_m|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \Rightarrow t = \frac{\hbar}{|E + E|} \cos^{-1}(-1) = \frac{\hbar}{2E} \cdot \pi \quad t = \frac{\pi\hbar}{2E}$$

(iii) Now for $|\psi(t)|^2 = |\psi(0)|^2$ we have

$$t = \frac{2\pi\hbar}{|E_n - E_m|} = \frac{2\pi\hbar}{2E} = \frac{\pi\hbar}{E} \Rightarrow \Delta t = t = \frac{\pi\hbar}{E}$$

(iv) We have $|\psi(0)\rangle = \frac{1}{\sqrt{2}}|\psi_+\rangle + \frac{1}{\sqrt{2}}|\psi_-\rangle$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} e^{\frac{iE_+t}{\hbar}} |\psi_+\rangle + \frac{1}{\sqrt{2}} e^{\frac{iE_-t}{\hbar}} |\psi_-\rangle$$

Now for $E_n = E, E_m = -E$ and $t = \frac{\hbar}{6E}$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} e^{\frac{iE}{\hbar} \cdot \frac{\hbar}{6E}} |\psi_+\rangle + \frac{1}{\sqrt{2}} e^{+\frac{iE}{\hbar} \cdot \frac{\hbar}{6E}} |\psi_-\rangle$$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \exp \frac{i\pi}{3} |\psi_+\rangle + \frac{1}{\sqrt{2}} \exp -\frac{i\pi}{3} |\psi_-\rangle$$

(v) At $t = \frac{\hbar}{6E}$ we have to find $\langle H \rangle$ and $\langle H^2 \rangle$

$$\langle H \rangle = \frac{\langle \psi(t) | H | \psi(t) \rangle}{\langle \psi(t) | \psi(t) \rangle} = \sum_n a_n P(a_n)$$

Here the eigen value $a_1 = E$ and $a_2 = -E$

$$\langle H \rangle = a_1 P(a_1) + a_2 P(a_2)$$

$$\langle H \rangle = EP(E) - EP(-E) = E \cdot \frac{1}{2} - E \cdot \frac{1}{2} = 0$$

$$\langle H^2 \rangle = \frac{\langle \psi | H^2 | \psi \rangle}{\langle \psi | \psi \rangle} = \sum_n a_n^2 P(a_n) = E^2 P(E) - (E)^2 P(-E) = \frac{E^2}{2} + \frac{E^2}{2} = E^2$$

$$\Delta H = \Delta E = \sqrt{\langle H^2 \rangle - \langle H \rangle^2} = \sqrt{E^2 - 0} = E$$

$$\Delta H \cdot \Delta t = E \cdot \frac{\pi\hbar}{E} = \pi\hbar$$

(vi) The probability that the system will be in the state at $t = \frac{h}{6E}$

$$P = \frac{|\langle \psi(t) | \psi(0) \rangle|^2}{\langle \psi(0) | \psi(0) \rangle}$$

$$\langle \psi(t) | = \frac{1}{\sqrt{2}} \exp \frac{-i\pi}{3} \langle \psi_+ | + \frac{1}{\sqrt{2}} \exp \frac{i\pi}{3} \langle \psi_- |$$

Probability P that state $|\psi(t)\rangle$ is in direction of state $|\psi(0)\rangle = \frac{|\langle \psi(t) | \psi(0) \rangle|^2}{\langle \psi(0) | \psi(0) \rangle}$

where $\langle \psi(t) | \psi(0) \rangle = \left| \frac{1}{\sqrt{2}} \right|^2 \exp \frac{i\pi}{3} + \left| \frac{1}{\sqrt{2}} \right|^2 \exp \frac{-i\pi}{3}$

$$P = \frac{\left| \frac{1}{2} \left(\exp \frac{i\pi}{3} + \exp \frac{-i\pi}{3} \right) \right|^2}{1} = \left| \frac{1}{2} \cdot 2 \cos \frac{\pi}{3} \right|^2 \Rightarrow P = \frac{1}{4}$$

Example: Consider a one-dimensional particle which is confined within the region

$0 \leq x \leq a$ and whose wave function is $\psi(x, t) = \sin\left(\frac{\pi x}{a}\right) e^{i\omega t}$. Find the potential $V(x)$.

Solution: From the fifth postulate.

$$H\psi = i\hbar \frac{\partial \psi}{\partial t} \quad H = \frac{P^2}{2m} + V(x) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi = i\hbar \frac{\partial \psi}{\partial t}$$

$$\frac{\pi^2 \hbar^2}{2ma^2} \sin \frac{\pi x}{a} e^{i\omega t} + V(x) \frac{\sin \pi x}{a} e^{i\omega t} = i\hbar \sin \frac{\pi x}{a} (-i\omega) e^{i\omega t}$$

$$\frac{\pi^2 \hbar^2}{2ma^2} + V(x) = -\hbar\omega \quad V(x) = -\hbar\omega - \frac{\pi^2 \hbar^2}{2ma^2} \quad V(x) = -\left(\hbar\omega + \frac{\pi^2 \hbar^2}{2ma^2} \right)$$